Final Technical Report

Network Design, Implementation and Verification

Molema Citrus Estates (Rustenburg)

Industry: Agriculture

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Student Number	38041693
Client ID	CLI-037
Project ID	CMPG325-2026-037
Assigned Address Block	192.168.26.0/24
Assigned Networking Feature	ACLs (traffic-filtering policy)
Design Constraint	VoIP traffic separated from data traffic
Change Request	CR14 — limited wireless access for after-hours cleaning/security contractors
Implementation Platform	Cisco Packet Tracer (Molema_Citrus_Phase2.pkt)
Final Submission	16 October 2026

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1. Executive Summary

This document outlines the design, execution, and validation of a segmented Layer 3 network for Molema Citrus Estates (Rustenburg), Client ID CLI-037, Project ID CMPG325-2026-037. Phase 1 employed VLSM to partition the designated 192.168.26.0/24 block into five VLAN subnets and a routed /30 link. Phase 2 executed the design within Cisco Packet Tracer. The report differentiates confirmed outcomes from functionalities that were not executed or could not be evaluated.

The configured network comprises Router0 serving as the edge router, a Cisco 3560 multilayer switch (Core Switch) functioning as the Layer 3 gateway and DHCP server, two Cisco 2960 access switches (AdminSwitch and Packhouse Switch), a wireless access point, and corresponding end devices. Five switch virtual interfaces (SVIs), two 802.1Q trunks, four DHCP pools, and static routing between Router0 and CoreSwitch were set up.

The designated function, ACL traffic filtering, was executed with two extended ACLs: GUEST_ISOLATION inbound on VLAN 50 and VOICE_ISOLATION inbound on VLAN 40. The report documents successful DHCP leases in Administration, Packhouse, Voice, and Guest, allowed gateway/inter-VLAN tests, and restricted tests from Guest and Voice to secure VLANs. Consider these results as confirmed only when the related endpoint screenshots and ACL counters are available and legible in the final report.

A significant limitation is clear: the Packet Tracer topology lacks an upstream ISP or Cloud device, and there is no implementation of NAT/PAT or a default route to an external network. Internet reachability (including 8.8.8.8) is consequently NOT CONFIRMED and should not be stated as successful.

2. Introduction and Client Requirements

2.1 Client background

Molema Citrus Estates is an agricultural organisation located in Rustenburg, North West Province, operating in the citrus industry. The project was assigned under Client ID CLI-037 and Project ID CMPG325-2026-037. The solution was required to remain specific to this client, and the address block 192.168.26.0/24 was allocated for all addressing.

2.2 Confirmed client requirements

The requirements below were taken directly from the project brief. Information not stated in the brief (for example, department headcounts) was treated as a design assumption in Phase 1 and was used only to size subnets.

Table 1: Client requirements and their implementation

Requirement	Source	Implementation (Phase 2)
Provide appropriate connectivity and network services for the assigned scenario	Brief, Section 6	Hierarchical topology with VLANs, DHCP and inter-VLAN routing (Section 4)
Use 192.168.26.0/24 as the basis of the IP addressing plan	Brief, Sections 6–7	VLSM allocation across five VLANs and one /30 link (Section 3.2)
VoIP traffic must be separated from data traffic	Brief, Section 8	Dedicated Voice VLAN 40, voice VLAN on phone ports, VOICE_ISOLATION ACL
Configure, verify and demonstrate ACLs (traffic-filtering policy)	Brief, Section 9	GUEST_ISOLATION and VOICE_ISOLATION (Section 5); verification (Section 6)
CR14: limited wireless access for after-hours cleaning/security contractors	Brief, Section 10	Isolated Guest VLAN 50 with GUEST_ISOLATION permitting DHCP and non-internal destinations only
Produce a working, testable Packet Tracer implementation	Brief, Section 6	Molema_Citrus_Phase2.pkt; tests documented in Section 6

2.3 Scope of the implementation

Phase 2 included cabling and setting up devices, creation of VLANs/SVIs, 802.1Q trunking, DHCP configuration, static routing between CoreSwitch and Router0, implementation of ACLs, and testing of endpoints. NAT/PAT, upstream Internet connectivity, and Internet reachability testing were unfinished; these constraints are discussed in Section 6.5.

3. Network Design (Phase 1 Summary)

3.1 Topology

A single-site estate was designed with a two-level hierarchical topology. Router0 functions as the boundary router. A Layer 3 core switch (CoreSwitch) enables inter-VLAN routing via SVIs and implements internal ACLs. Two access switches, one designated for the Administration/Office section (AdminSwitch) and another for the Packhouse/Operations section (PackhouseSwitch), link user and voice endpoints, while a wireless access point enables contractor access on the Guest VLAN. A management workstation is connected directly to the core. Figures 1 and 2 illustrate the physical and logical topologies of Phase 1.

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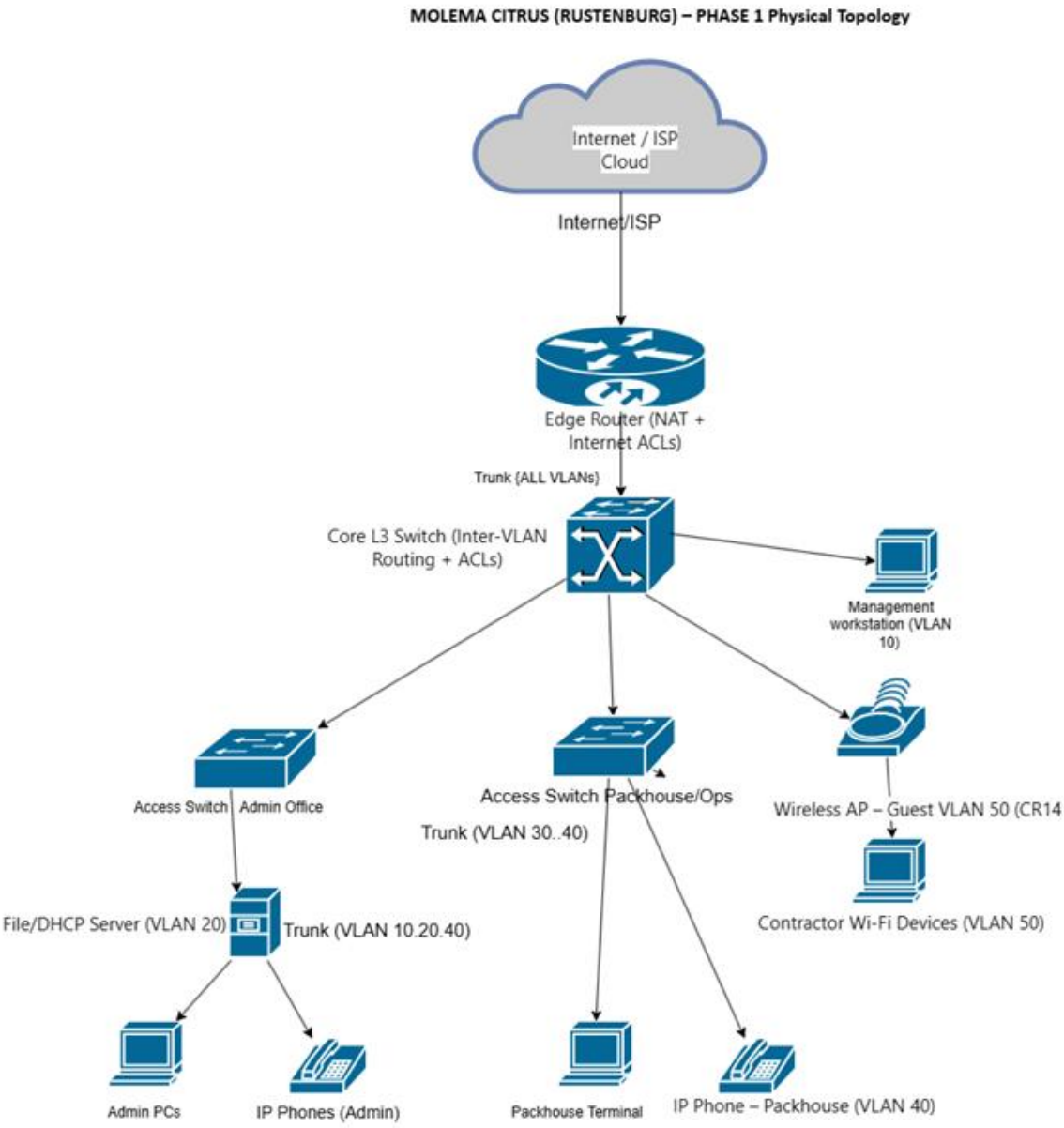


Figure 1: Phase 1 physical topology (design)

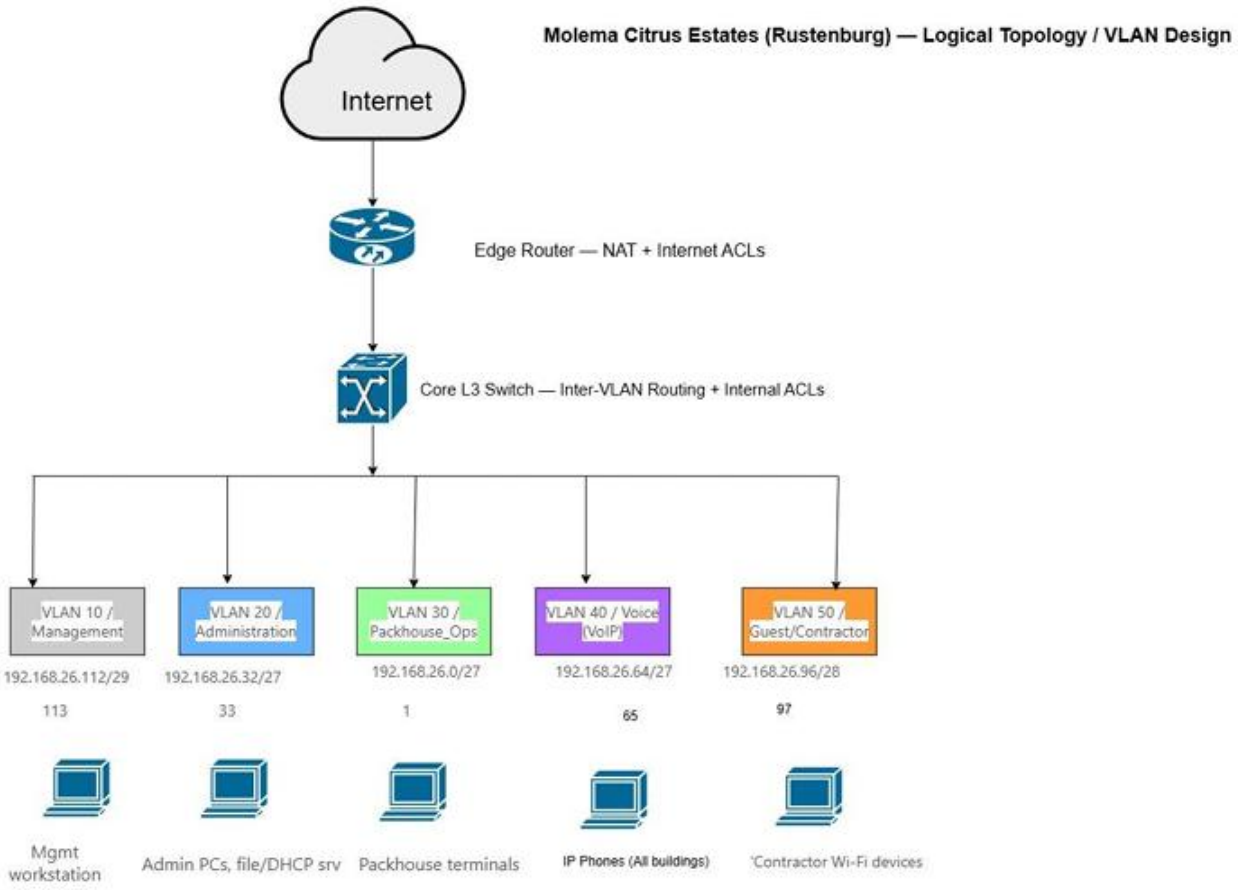


Figure 2: Phase 1 logical topology and VLAN design

3.2 IP addressing plan

The block 192.168.26.0/24 (256 addresses) was split using VLSM, with segments assigned in descending order based on needed host count. Host counts were design estimates (Administration ≈ 20, Packhouse ≈ 30, Voice ≈ 15, Guest ≈ 10, Management ≈ 5) utilized only for subnet sizing. A /28 offers 14 usable addresses ($2^4 - 2$) and thus cannot support 20 or 15 hosts; a /27 (with 30 usable addresses) was chosen for the Administration and Voice segments. The unused range, 192.168.26.124 to 192.168.26.255 (132 addresses), was kept unallocated and set aside for future expansion

Table 2: Phase 1 VLSM allocation

Network	Hosts	Subnet	Mask	Usable range	Broadcast	Gateway
Packhouse (VLAN 30)	30	192.168.26.0/27	255.255.255.224	.1 – .30	192.168.26.31	192.168.26.1
Administration (VLAN 20)	20	192.168.26.32/27	255.255.255.224	.33 – .62	192.168.26.63	192.168.26.33
Voice (VLAN 40)	15	192.168.26.64/27	255.255.255.224	.65 – .94	192.168.26.95	192.168.26.65
Guest (VLAN 50)	10	192.168.26.96/28	255.255.255.240	.97 – .110	192.168.26.111	192.168.26.97
Management (VLAN 10)	5	192.168.26.112/29	255.255.255.248	.113 – .118	192.168.26.119	192.168.26.113
Point-to-point link	2	192.168.26.120/30	255.255.255.252	.121 – .122	192.168.26.123	n/a

3.3 VLAN structure

Table 3: VLAN plan

VLAN	Name	Purpose	Subnet
10	Management	Administrative access to network devices	192.168.26.112/29
20	Administration	Office and administrative devices	192.168.26.32/27
30	Packhouse	Packhouse and operations terminals	192.168.26.0/27
40	Voice	IP telephony endpoints (all buildings)	192.168.26.64/27
50	Guest	After-hours contractor access (CR14)	192.168.26.96/28

3.4 Design justification

A hierarchical structure with a specific core was chosen because the client needs to manage multiple distinct traffic classes (data, voice, guest, and management) that necessitate routing and policies among them; a flat network cannot fulfill the VoIP separation requirement or the guest isolation need of CR14. Five VLANs were implemented to ensure each traffic class creates its own broadcast domain and policy boundary. VLSM was chosen instead of equal-sized subnetting because it aligns each subnet with its real needs, prevents the allocation of large blocks to smaller segments like Management, and retains over half of the assigned block for future expansion. Static and connected routing was deemed adequate for a single location with one exit point; implementing a dynamic routing protocol would have introduced unnecessary complexity without meeting any specified needs. The CoreSwitch was assigned as the Layer 3 gateway to centralize routing and ACL enforcement within one device.

4. Implementation (Phase 2)

4.1 Platform and device configuration overview

The network was created in Cisco Packet Tracer and stored as Molema_Citrus_Phase2.pkt. Router0 (Cisco 2911) was implemented as the edge router, while a Cisco 3560-24PS multilayer switch (CoreSwitch) served as the core. Two Cisco 2960-24TT switches (AdminSwitch and PackhouseSwitch) were installed at the access level. All connections between devices utilized copper straight-through cables. Figure 3 illustrates the finished topology.

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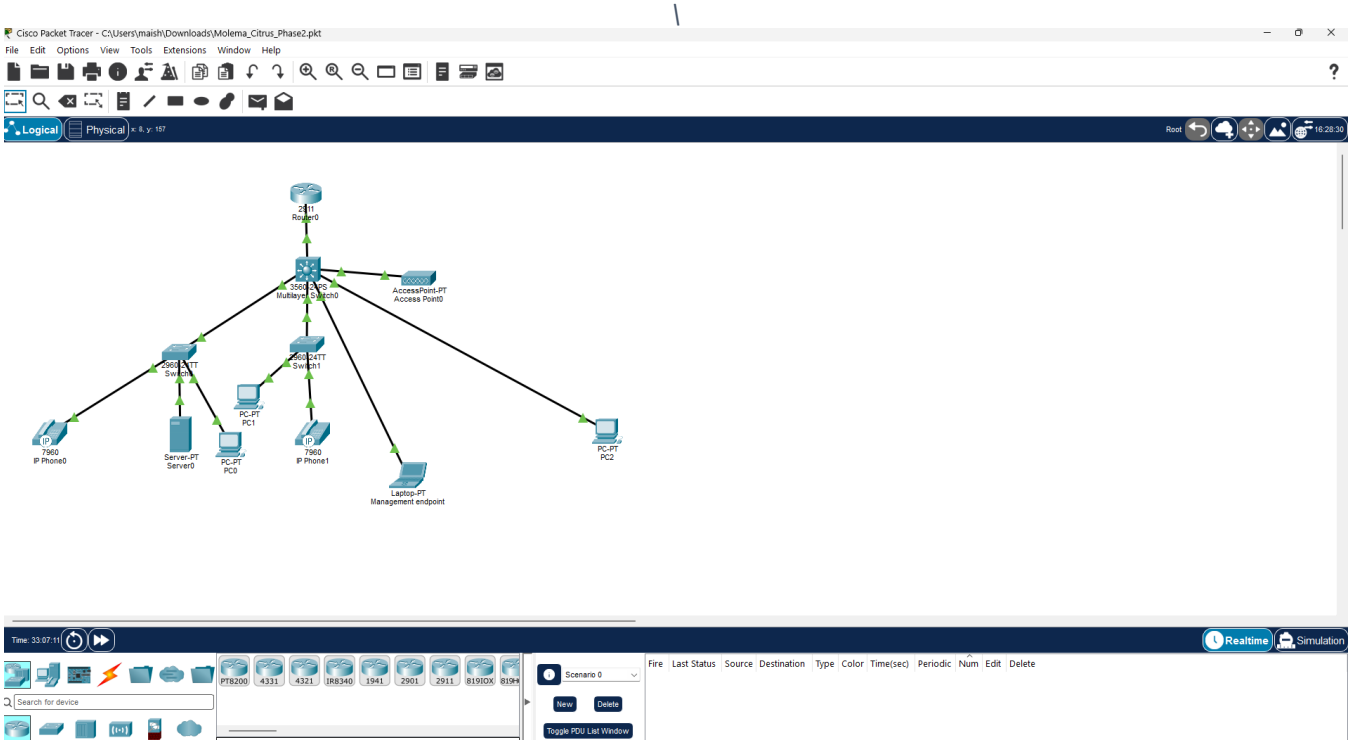


Figure 3: Completed Molema Citrus Estates Packet Tracer topology

Table 4: Implemented devices and roles

Device	Model / type	Role
Router0	Cisco 2911	Edge router; routed link to CoreSwitch (Gi0/0, 192.168.26.121/30)
CoreSwitch	Cisco 3560-24PS	Layer 3 gateway (SVIs), DHCP server, static routing, trunks, Guest/Voice ACLs
AdminSwitch	Cisco 2960-24TT	Access switch for Administration, Management and Voice traffic
PackhouseSwitch	Cisco 2960-24TT	Access switch for Packhouse and Voice traffic
AccessPoint0	Access Point-PT	Wireless access component, cabled to CoreSwitch
Server0	Server-PT	Server endpoint with a static IPv4 address in VLAN 20
PC0, PC1, Guest PC	PC-PT	Client endpoints used to verify DHCP, routing and ACL behaviour
IP Phone0, IP Phone1	Cisco 7960	Voice endpoints on VLAN 40
Laptop0	Laptop-PT	Management endpoint (VLAN 10, static addressing)

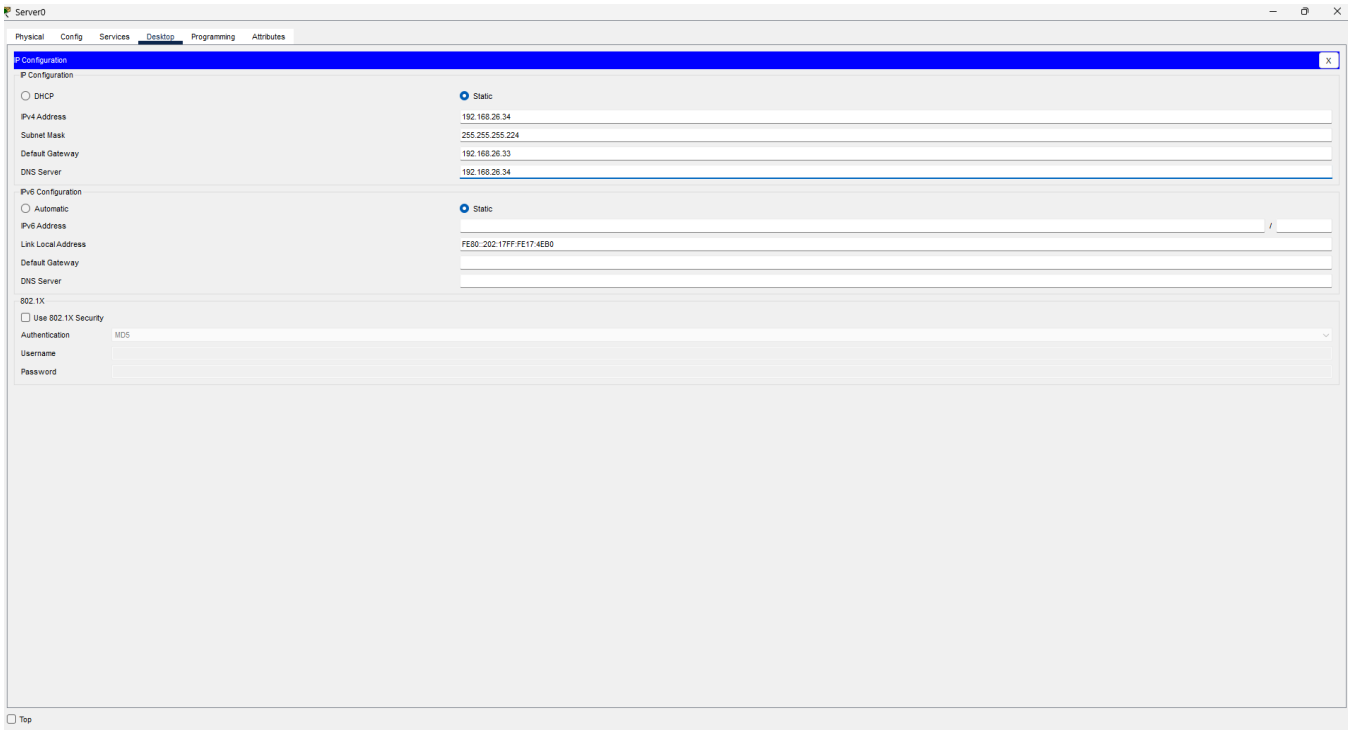


Figure 4: Server0 static IP configuration

4.2 VLAN and SVI configuration

The CoreSwitch had five VLANs established and global IP routing was activated. Every VLAN was given an SVI with its address being the first usable address of the subnet, allowing the CoreSwitch to serve as the default gateway for all VLANs. The practice of designating the first available address as gateway was upheld uniformly.

```
ip routing
!
vlan 10
 name Management
vlan 20
 name Admin
vlan 30
 name Packhouse
vlan 40
 name Voice
vlan 50
 name Guest
!
interface Vlan10
 ip address 192.168.26.113 255.255.255.248
 no shutdown
interface Vlan20
 ip address 192.168.26.33 255.255.255.224
 no shutdown
interface Vlan30
 ip address 192.168.26.1 255.255.255.224
 no shutdown
interface Vlan40
 ip address 192.168.26.65 255.255.255.224
 no shutdown
interface Vlan50
 ip address 192.168.26.97 255.255.255.240
  no shutdown
```

The command show ip interface brief was used to confirm that all five SVIs were in the up/up state (Figure 5).

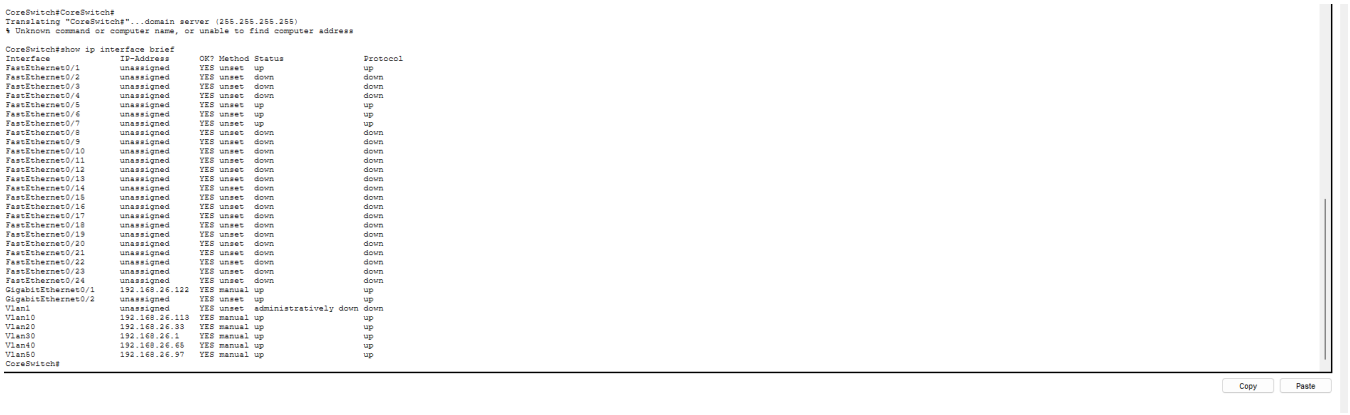


Figure 5: SVI verification: all five VLAN interfaces up/up

4.3 Access-switch and voice VLAN configuration

The access switches were labeled AdminSwitch and PackhouseSwitch. Ports linked to IP phones were set up with a data VLAN as well as a voice VLAN. The data VLAN (20 on AdminSwitch, 30 on PackhouseSwitch) transports untagged traffic from a PC linked to the phone's pass-through port, while the voice VLAN (40) is communicated to the phone, which subsequently tags its own traffic using 802.1Q. Thus, voice and data traffic utilize a single cable while staying on distinct VLANs, fulfilling the VoIP-separation requirement at Layer 2. Internet reachability (including 8.8.8.8) is thus NOT CONFIRMED and should not be communicated as successful.

```
! AdminSwitch
hostname AdminSwitch
interface FastEthernet0/3
 switchport access vlan 20
 switchport voice vlan 40

! PackhouseSwitch
hostname PackhouseSwitch
interface FastEthernet0/2
 switchport access vlan 30
 switchport voice vlan 40
```

```
PackhouseSwitch#show interfaces fastEthernet 0/2 switchport
Name: Fa0/2
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 30 (VLAN0030)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: 40
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none


PackhouseSwitch#
```

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Figure 6: Packhouse IP phone port configuration: data VLAN 30 and voice VLAN 40

On the CoreSwitch, Laptop0 was connected to FastEthernet0/5 as an access port in VLAN 10, and a Guest test endpoint was connected to FastEthernet0/7 as an access port in VLAN 50.

```
Interface FastEthernet0/5
 switchport mode access
 switchport access vlan 10
interface FastEthernet0/7
 switchport mode access
 switchport access vlan 50
```

4.4 Trunking (802.1Q)

Two 802.1Q trunks were set up on the CoreSwitch. Gi0/2 is linked to AdminSwitch and transmits VLANs 10, 20, and 40; Fa0/1 is connected to PackhouseSwitch and transmits VLANs 30 and 40. The allowed-VLAN list on each trunk was limited to the VLANs needed by the connected access area, which reduces unnecessary broadcast traffic and minimizes VLAN exposure to areas that don't need them. On the Cisco 3560, the trunk encapsulation needed to be defined explicitly prior to accepting trunk mode (Section 6.4).

```
interface GigabitEthernet0/2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,40
interface FastEthernet0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 30,40
```

Table 5: Verified 802.1Q trunks on CoreSwitch

Core port	Connected area	Allowed VLANs	Status
Gi0/2	AdminSwitch	10, 20, 40	Trunking (802.1Q)
Fa0/1	PackhouseSwitch	30, 40	Trunking (802.1Q)

```
CoreSwitch# show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Gig0/2      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       30,40
Gig0/2      10,20,40

Port        Vlans allowed and active in management domain
Fa0/1       30,40
Gig0/2      10,20,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       30,40
Gig0/2      10,20,40

CoreSwitch#
```

Figure 7: CoreSwitch trunk VLAN verification (show interfaces trunk)

4.5 DHCP configuration

DHCP was set up directly on the CoreSwitch with four IOS pools, one designated for Administration, Packhouse, Voice, and Guest networks. Since the CoreSwitch serves as the gateway for all VLANs, DHCP broadcasts are captured on the related SVIs, negating the need for relay (ip helper-address) setup. Management devices (VLAN 10) are designed to use static addressing, so no pool was established for that VLAN. Gateway and reserved infrastructure addresses were not included in dynamic allocation. The Voice pool also includes DHCP option 150, which provides the address of the reference for voice services (TFTP) utilized by IP phones.

```
ip dhcp excluded-address 192.168.26.1 192.168.26.4
ip dhcp excluded-address 192.168.26.33 192.168.26.39
ip dhcp excluded-address 192.168.26.65 192.168.26.69
ip dhcp excluded-address 192.168.26.97 192.168.26.99
!
ip dhcp pool ADMIN_POOL
 network 192.168.26.32 255.255.255.224
 default-router 192.168.26.33
 dns-server 8.8.8.8
ip dhcp pool PACKHOUSE_POOL
 network 192.168.26.0 255.255.255.224
 default-router 192.168.26.1
 dns-server 8.8.8.8
ip dhcp pool VOICE_POOL
 network 192.168.26.64 255.255.255.224
 default-router 192.168.26.65
 dns-server 8.8.8.8
 option 150 ip 192.168.26.65
ip dhcp pool GUEST_POOL
 network 192.168.26.96 255.255.255.240
 default-router 192.168.26.97
 dns-server 8.8.8.8
```

Successful dynamic addressing was observed on all four networks. Examples recorded during testing included an Administration client at 192.168.26.40, Voice clients within 192.168.26.64/27, and a Guest client at 192.168.26.101 (Figure 8 to Figure 12).

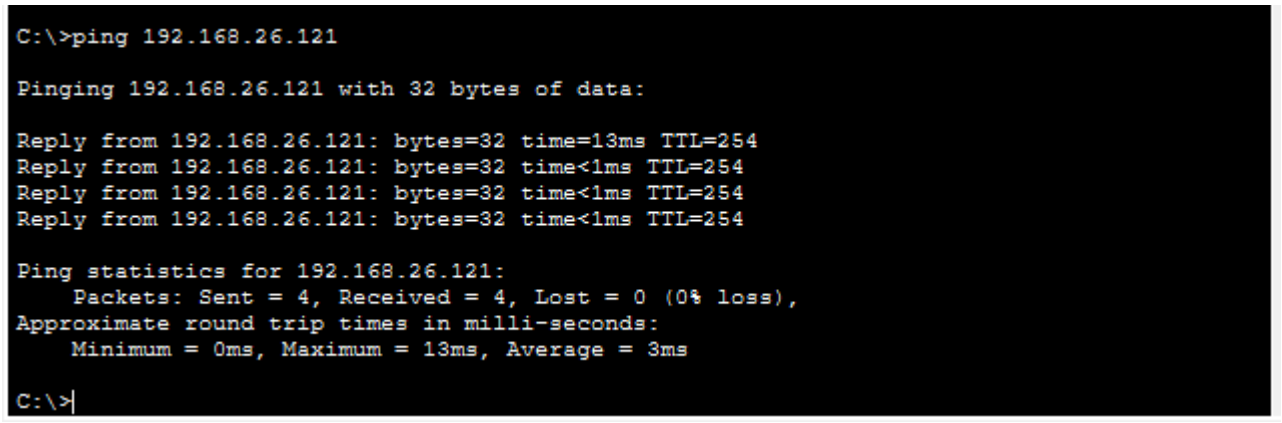


Figure 8: Administration VLAN 20 DHCP assignment on PC0

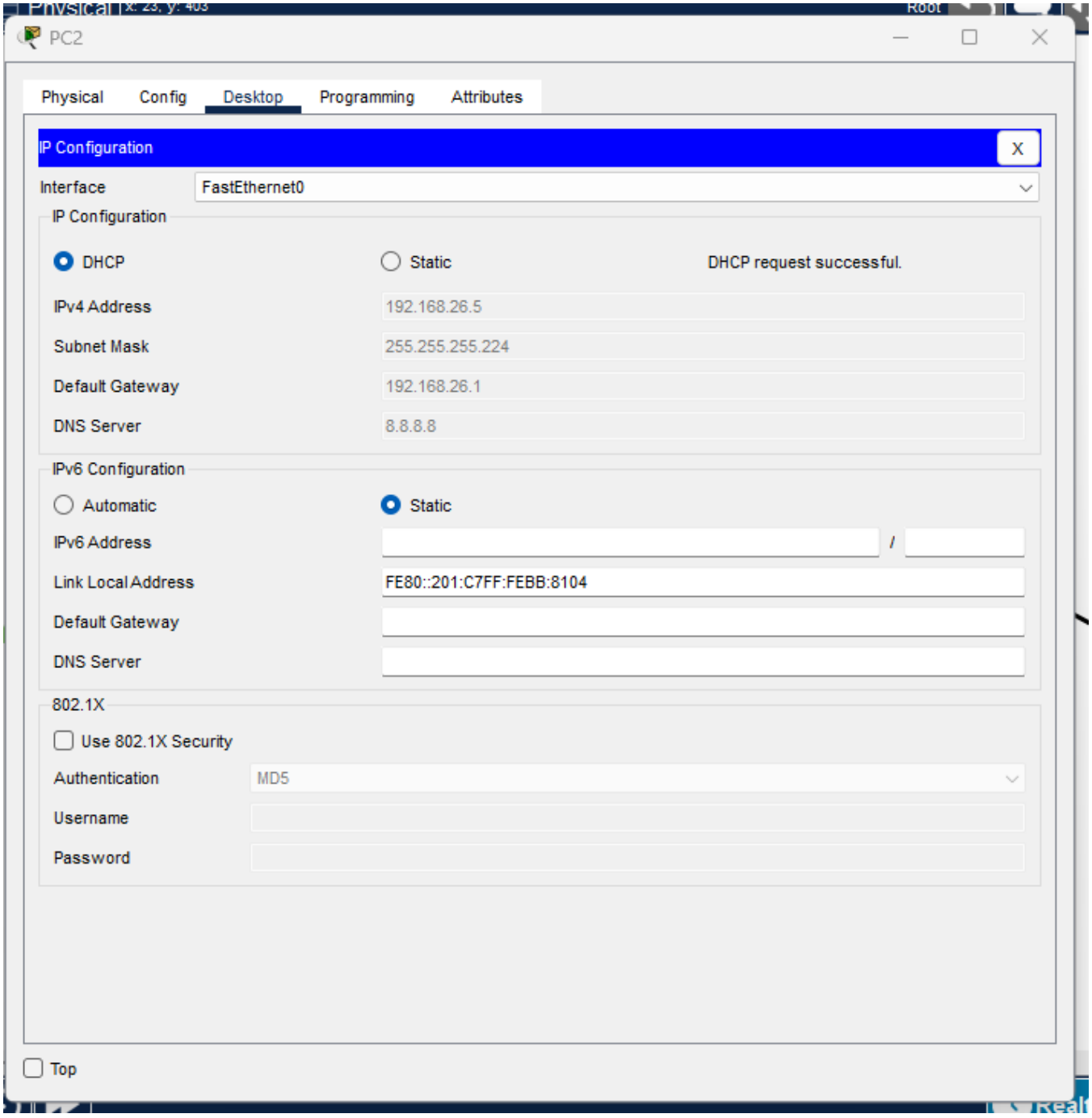


Figure 9: Packhouse VLAN 30 successful DHCP address assignment

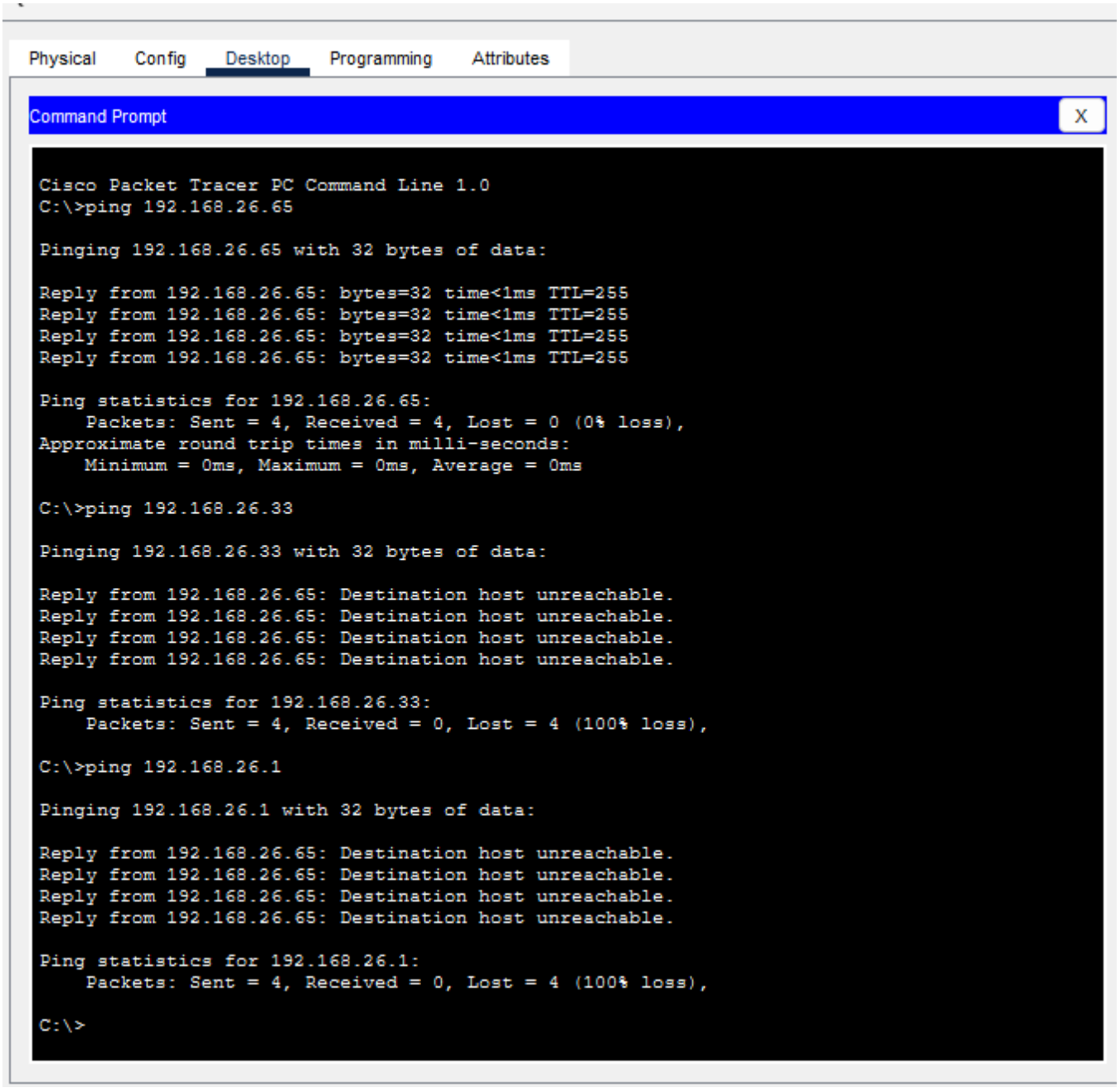


Figure 10: Voice VLAN 40 DHCP assignment to IP phones

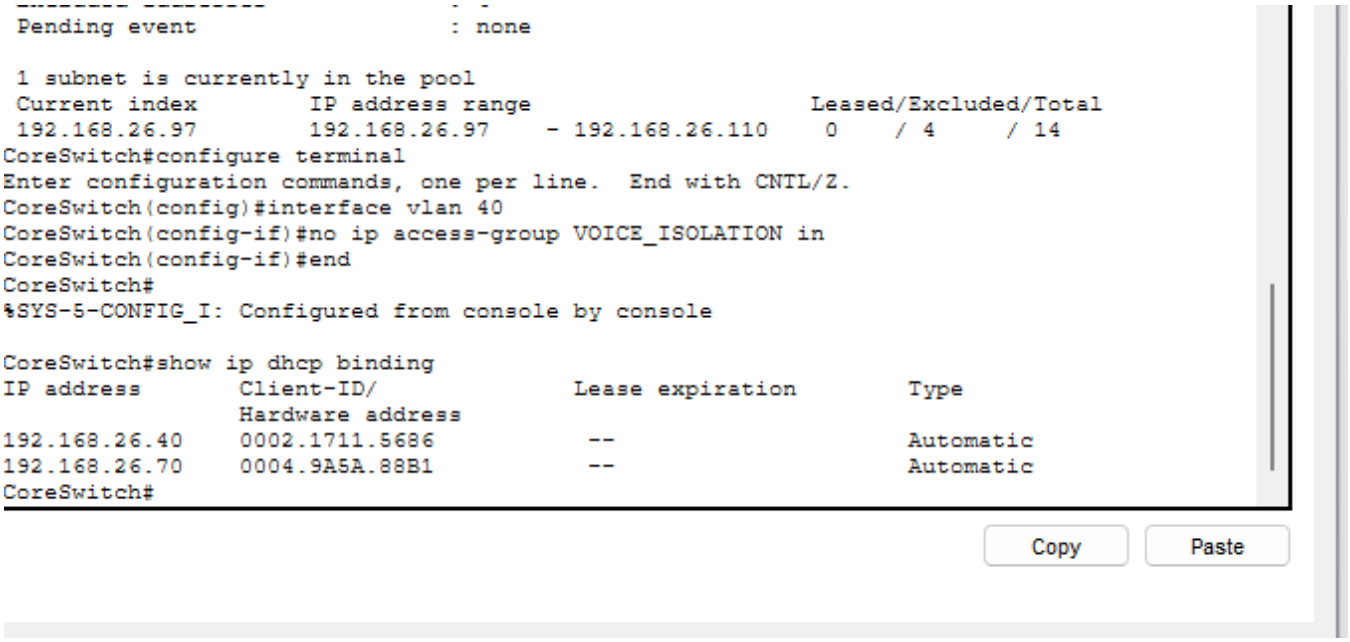


Figure 11: Voice VLAN 40 DHCP verification with VOICE_ISOLATION ACL active

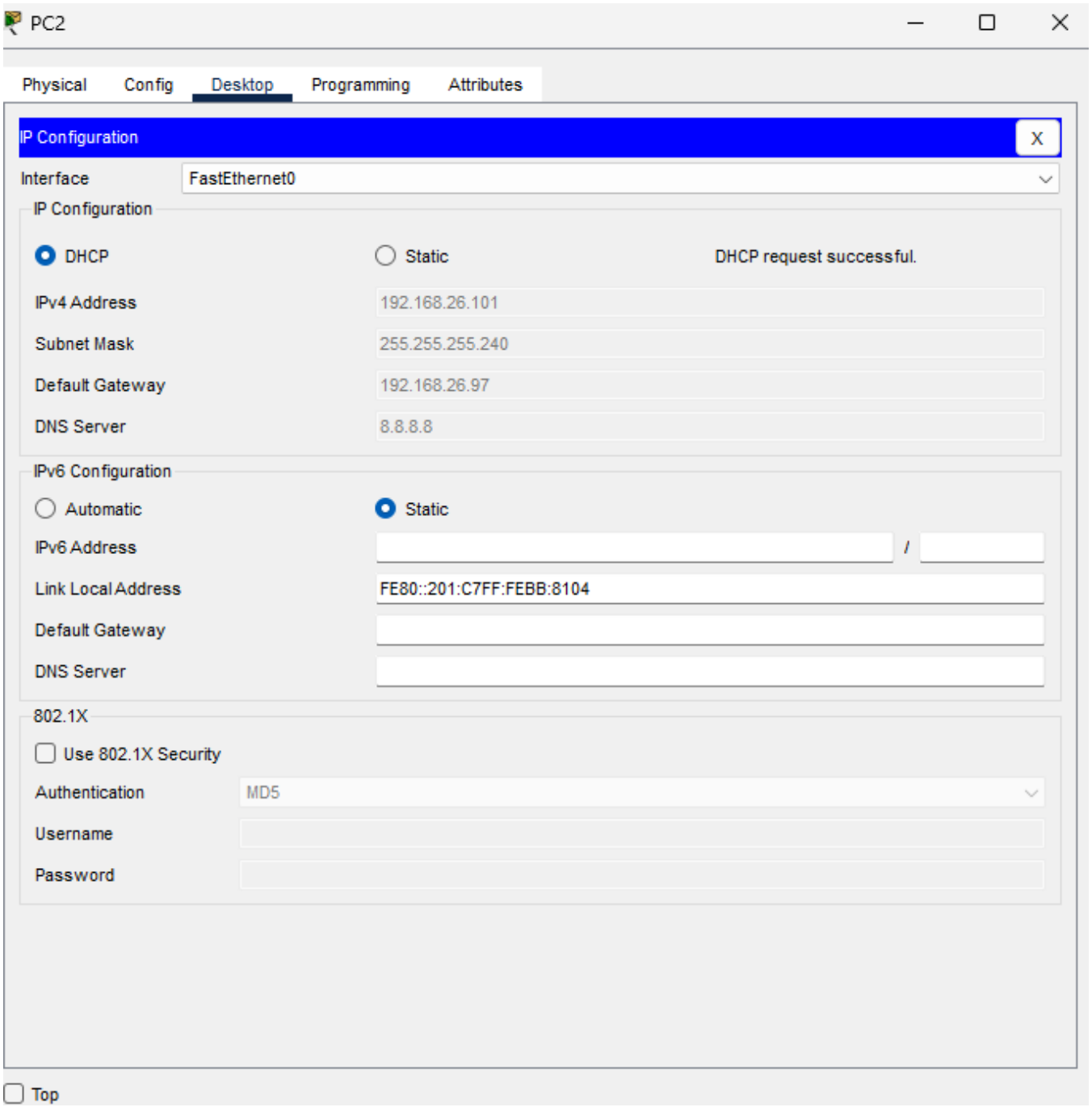


Figure 12: Guest VLAN 50 successful DHCP address assignment

4.6 Routing configuration

Inter-VLAN routing is performed by the CoreSwitch through its five SVIs, which are directly connected routes. The CoreSwitch is connected to Router0 by a routed point-to-point link: Gi0/1 on the CoreSwitch was converted to a routed port (no switchport) with address 192.168.26.122/30, and Router0 Gi0/0 was configured with 192.168.26.121/30. Router0 was configured with static routes toward every internal VLAN subnet, using the CoreSwitch address 192.168.26.122 as next hop. Reachability across the link was confirmed by a successful ping from an Administration client to Router0 (Figure 13).

```
! CoreSwitch
interface GigabitEthernet0/1
 no switchport
 ip address 192.168.26.122 255.255.255.252
 no shutdown

! Router0
interface GigabitEthernet0/0
 ip address 192.168.26.121 255.255.255.252
 no shutdown
ip route 192.168.26.0 255.255.255.224 192.168.26.122
ip route 192.168.26.32 255.255.255.224 192.168.26.122
ip route 192.168.26.64 255.255.255.224 192.168.26.122
ip route 192.168.26.96 255.255.255.240 192.168.26.122
ip route 192.168.26.112 255.255.255.248 192.168.26.122
```

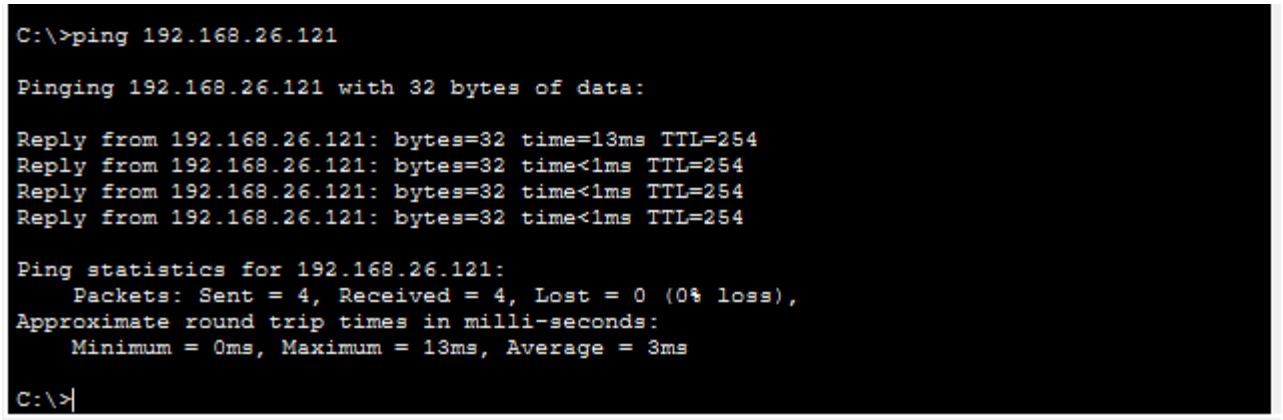


Figure 13: Administration VLAN 20 to Router0 connectivity test: PASS

4.7 Deviations from the Phase 1 design

The implementation followed the approved Phase 1 design with the following documented differences, each arising from the constraints of the Packet Tracer environment.

Table 6: Deviations from Phase 1

Phase 1 design	Phase 2 implementation	Reason
DHCP service hosted on the Administration file/DHCP server	DHCP pools hosted on CoreSwitch; Server0 DHCP service disabled	Leases from the server-based service were returned without default-gateway and DNS values during testing; IOS pools on the CoreSwitch returned complete leases
192.168.26.120/30 reserved for the router-to-ISP link	Same /30 used for the CoreSwitch-to-Router0 routed link (.121 / .122)	No ISP or Cloud device was implemented in the topology
NAT/PAT overload and Internet-facing ACLs on the edge router	Not configured	No outside interface or upstream Internet path exists in the topology
Contractor access via wireless AP on VLAN 50	Guest-VLAN behaviour verified using a test endpoint in VLAN 50 on CoreSwitch Fa0/7	Provides a repeatable endpoint from which ACL behaviour could be measured

5. Assigned Feature: ACL Implementation

5.1 Objective

The assigned networking feature was traffic filtering using access-control lists. The objective was to isolate the Guest VLAN (VLAN 50) and the Voice VLAN (VLAN 40) from the protected internal VLANs, thereby satisfying the CR14 requirement for limited contractor access and the constraint that VoIP traffic be separated from data traffic, while still permitting the traffic that the design intentionally allows (DHCP and the endpoints' own gateways).

5.2 ACL configuration

Two named extended IPv4 ACLs were created on the CoreSwitch. Named ACLs were used in preference to numbered ACLs because the name states the purpose of the policy directly in the configuration.

GUEST_ISOLATION (applied inbound on VLAN 50)

```
ip access-list extended GUEST_ISOLATION
 permit udp any eq bootpc any eq bootps
 permit udp any eq bootps any eq bootpc
 deny ip 192.168.26.96 0.0.0.15 192.168.26.0 0.0.0.31
 deny ip 192.168.26.96 0.0.0.15 192.168.26.32 0.0.0.31
 deny ip 192.168.26.96 0.0.0.15 192.168.26.64 0.0.0.31
 deny ip 192.168.26.96 0.0.0.15 192.168.26.112 0.0.0.7
 permit ip 192.168.26.96 0.0.0.15 any
!
interface Vlan50
 ip access-group GUEST_ISOLATION in
```

VOICE_ISOLATION (applied inbound on VLAN 40)

```
ip access-list extended VOICE_ISOLATION
 permit udp any eq bootpc any eq bootps
 permit udp any eq bootps any eq bootpc
 deny ip 192.168.26.64 0.0.0.31 192.168.26.0 0.0.0.31
 deny ip 192.168.26.64 0.0.0.31 192.168.26.32 0.0.0.31
 deny ip 192.168.26.64 0.0.0.31 192.168.26.112 0.0.0.7
 deny ip 192.168.26.64 0.0.0.31 192.168.26.96 0.0.0.15
 permit ip 192.168.26.64 0.0.0.31 any
!
interface Vlan40
 ip access-group VOICE_ISOLATION in
```

The entries are processed top-down, and processing stops at the first matching ACE, so rule order is significant. The DHCP permits are intended to allow client/server DHCP traffic before the isolation rules; confirm the behaviour using the corresponding lease evidence and ACL counters.

- The two UDP entries (BOOTP/DHCP client and server ports) are placed first so that DHCP exchanges are permitted before any isolation rule is evaluated.
- The deny entries block traffic sourced from the Guest (or Voice) subnet toward the Packhouse (192.168.26.0/27), Administration (192.168.26.32/27), Management (192.168.26.112/29) and, as applicable, Voice or Guest networks.
- The final permit entry allows other IPv4 destinations from the source subnet unless an earlier deny matches. This does not by itself prove Internet access: forwarding still requires a valid route and an operational external path. Without the final permit, the implicit "deny ip any any" would discard all remaining IPv4 traffic.

The Management entries use the wildcard mask 0.0.0.7, which matches exactly the Management /29 (192.168.26.112 – 192.168.26.119). The adjacent point-to-point link (192.168.26.120/30) is therefore not covered by the isolation entries. An earlier version of these entries used 0.0.0.15, which also matched the router link; this was corrected and the tests repeated (Section 6.4).

5.3 Placement rationale

Both ACLs were applied inbound on the relevant SVIs of the CoreSwitch rather than on Router0, for the following reasons.

- Traffic between VLANs is routed by the CoreSwitch and never traverses Router0. An ACL on Router0 would therefore never see Guest-to-Administration or Voice-to-Packhouse traffic.
- Filtering inbound at the source SVI discards prohibited packets at the first Layer 3 interface they reach, before any routing lookup or forwarding across the internal network. This conserves forwarding resources and follows the principle of filtering as close to the source as possible.
- Because each ACL is evaluated only for packets entering the Guest or Voice SVI, return traffic of permitted sessions arrives on other interfaces and is not evaluated by the ACL. A stateless extended ACL was therefore sufficient, and reflexive or established-based filtering was not required.
- Internal segmentation policy (CoreSwitch) is kept separate from any Internet-facing policy (Router0), which keeps each configuration simpler to read and audit.

5.4 Troubleshooting: DHCP and ACL ordering

After the isolation ACLs were first applied, a client in a filtered VLAN failed to obtain a DHCP lease and was assigned an APIPA address (169.254.x.x). The cause was that the initial ACL did not accommodate DHCP. A DHCP DISCOVER message is sent with source address 0.0.0.0 and destination 255.255.255.255; it therefore matches none of the subnet-based entries and was discarded by the implicit deny at the end of the ACL.

The ACLs were rebuilt with the UDP BOOTP/DHCP permit entries placed ahead of the isolation rules (as shown in Section 5.2). After the correction, the Guest client received 192.168.26.101 and Voice DHCP bindings were observed with the ACL active. This confirmed that the isolation policy and DHCP service could operate together.

5.5 Verification of ACL operation

The command show access-lists was used to confirm that both ACLs were present with the intended entries and that match counters increased while the ping tests of Section 6 were performed. Non-zero counters on the deny entries provide direct evidence that the ACLs processed real packets and were responsible for the blocked results, rather than the failures arising from a routing or addressing fault (Figure 14).

```
CoreSwitch#%DHCPD-4-PING_CONFLICT: DHCP address conflict:  server pinged 192.168.26.100.

CoreSwitch#show access-lists GUEST_ISOLATION
Extended IP access list GUEST_ISOLATION
    permit udp any eq bootpc any eq bootps (6 match(es))
    permit udp any eq bootps any eq bootpc
    deny ip 192.168.26.96 0.0.0.15 192.168.26.0 0.0.0.31 (4 match(es))
    deny ip 192.168.26.96 0.0.0.15 192.168.26.32 0.0.0.31 (4 match(es))
    deny ip 192.168.26.96 0.0.0.15 192.168.26.64 0.0.0.31 (4 match(es))
    deny ip 192.168.26.96 0.0.0.15 192.168.26.112 0.0.0.15 (4 match(es))
    permit ip 192.168.26.96 0.0.0.15 any (7 match(es))

CoreSwitch#show access-lists VOICE_ISOLATION
Extended IP access list VOICE_ISOLATION
    permit udp any eq bootpc any eq bootps (4205 match(es))
    permit udp any eq bootps any eq bootpc
    deny ip 192.168.26.64 0.0.0.31 192.168.26.0 0.0.0.31 (4 match(es))
    deny ip 192.168.26.64 0.0.0.31 192.168.26.32 0.0.0.31 (4 match(es))
    deny ip 192.168.26.64 0.0.0.31 192.168.26.112 0.0.0.15 (4 match(es))
    deny ip 192.168.26.64 0.0.0.31 192.168.26.96 0.0.0.15 (4 match(es))
    permit ip 192.168.26.64 0.0.0.31 any (1407 match(es))

CoreSwitch#
```

Figure 14: ACL match-counter verification for Guest and Voice isolation (show access-lists)

6. Testing and Verification

6.1 Methodology

Testing was performed using ping tests issued from actual Packet Tracer endpoints (PCs, IP phones and the Guest test PC) rather than from the network devices themselves, so that the results reflect the experience of real clients. Each test was classified against an expected result. Where the security policy requires that communication be blocked, a failed ping is recorded as PASS,

because the observed behaviour matched the requirement; the table therefore distinguishes functional failures from successful isolation.

6.2 Results

Table 7: Test results

Test	Expected	Outcome	Interpretation / evidence
DHCP: Administration client	Lease obtained	**PASS**	192.168.26.40 assigned (Figure 8)
DHCP: Packhouse client	Lease obtained	**PASS**	Address in 192.168.26.0/27 (Figure 9)
DHCP: Voice endpoints	Lease obtained	**PASS**	Addresses in 192.168.26.64/27 (Figure 10, Figure 11)
DHCP: Guest client	Lease obtained	**PASS**	192.168.26.101 assigned (Figure 12)
Admin client → Admin gateway 192.168.26.33	Reachable	**PASS**	Gateway connectivity confirmed (Figure 15)
Admin → Packhouse	Reachable	**PASS**	Inter-VLAN routing confirmed (Figure 16)
Admin → Router0 192.168.26.121	Reachable	**PASS**	Core-to-edge routing confirmed (Figure 13)
Voice test host → Voice gateway 192.168.26.65	Reachable	**PASS**	Local gateway reachable (Figure 17)
Voice → Admin	Blocked	**PASS**	VOICE_ISOLATION deny matched (Figure 18)
Voice → Packhouse	Blocked	**PASS**	VOICE_ISOLATION deny matched (Figure 18)
Voice → Guest	Blocked	**PASS**	VOICE_ISOLATION deny matched (Figure 19)
Voice → Management	Blocked	**PASS**	VOICE_ISOLATION deny matched (Figure 20)
Guest → Admin	Blocked	**PASS**	GUEST_ISOLATION deny matched (Figure 21)
Guest → Packhouse	Blocked	**PASS**	GUEST_ISOLATION deny matched (Figure 21)
Guest → Voice	Blocked	**PASS**	GUEST_ISOLATION deny matched (Figure 21)
Guest → Management	Blocked	**PASS**	GUEST_ISOLATION deny matched (Figure 21)
Guest → 8.8.8.8 (Internet)	Internet path unavailable	**NOT VERIFIED**	No upstream path in topology; success is not claimed (Section 6.5)

6.3 Supporting evidence

Permitted-traffic tests are shown in Figure

```
Pinging 192.168.26.33 with 32 bytes of data:

Reply from 192.168.26.33: bytes=32 time=13ms TTL=255
Reply from 192.168.26.33: bytes=32 time<1ms TTL=255
Reply from 192.168.26.33: bytes=32 time<1ms TTL=255
Reply from 192.168.26.33: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.26.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>
```

Figure 15: Administration client to Administration gateway: PASS

```
C:\>ping 192.168.26.1

Pinging 192.168.26.1 with 32 bytes of data:

Reply from 192.168.26.1: bytes=32 time<1ms TTL=255
Reply from 192.168.26.1: bytes=32 time<1ms TTL=255
Reply from 192.168.26.1: bytes=32 time<1ms TTL=255
Reply from 192.168.26.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.26.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 16: Administration to Packhouse permitted inter-VLAN test: PASS

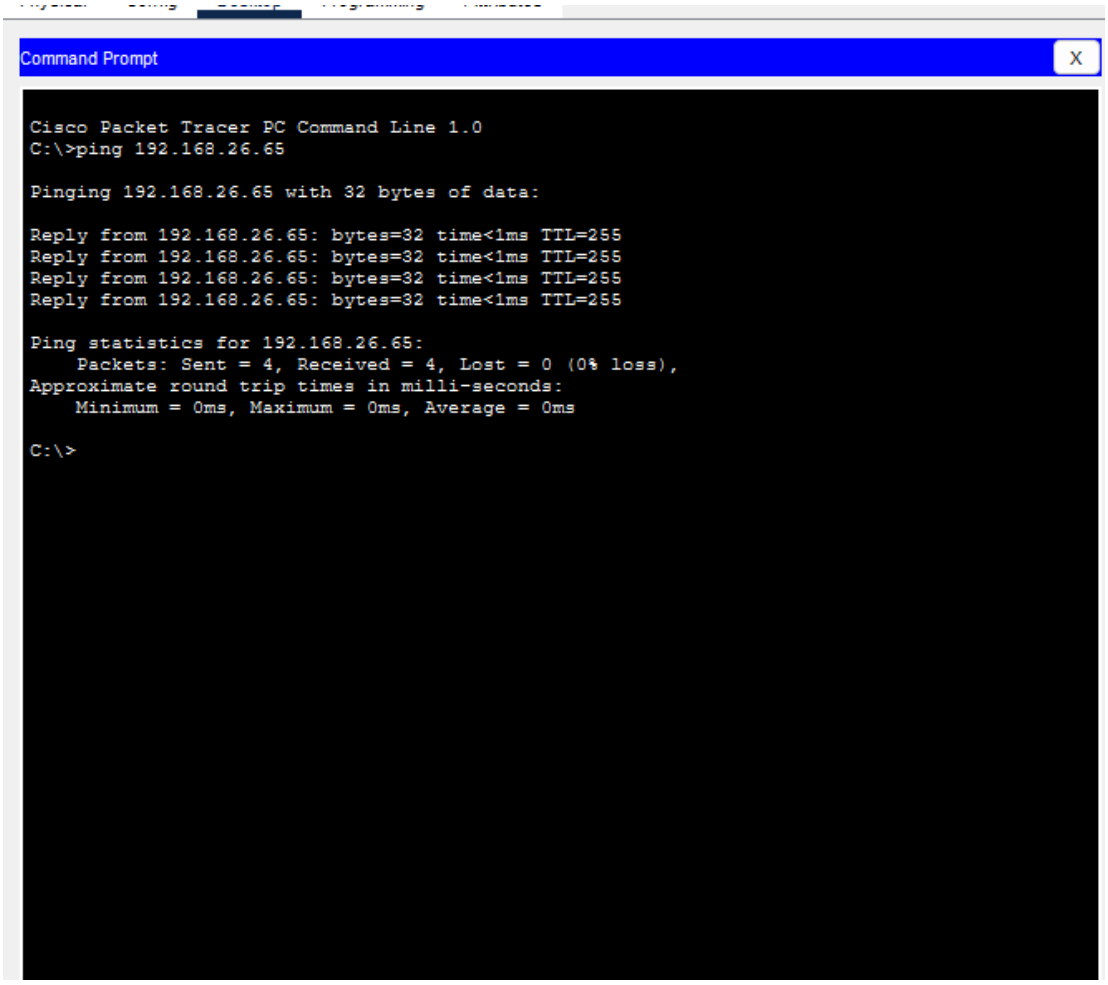


Figure 17: Voice VLAN 40 gateway connectivity test: successful

```
C:\>ping 192.168.26.97

Pinging 192.168.26.97 with 32 bytes of data:

Reply from 192.168.26.65: Destination host unreachable.
Reply from 192.168.26.65: Destination host unreachable.
Reply from 192.168.26.65: Destination host unreachable.
Reply from 192.168.26.65: Destination host unreachable.

Ping statistics for 192.168.26.97:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 18: Voice VLAN 40 isolation: Administration and Packhouse networks blocked as intended

```
C:\>ping 192.168.26.97

Pinging 192.168.26.97 with 32 bytes of data:

Reply from 192.168.26.65: Destination host unreachable.
Reply from 192.168.26.65: Destination host unreachable.
Reply from 192.168.26.65: Destination host unreachable.
Reply from 192.168.26.65: Destination host unreachable.

Ping statistics for 192.168.26.97:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 19: Voice VLAN 40 isolation test: Guest VLAN 50 blocked

```
C:\>ping 192.168.26.113

Pinging 192.168.26.113 with 32 bytes of data:

Reply from 192.168.26.65: Destination host unreachable.
Reply from 192.168.26.65: Destination host unreachable.
Reply from 192.168.26.65: Destination host unreachable.
Reply from 192.168.26.65: Destination host unreachable.

Ping statistics for 192.168.26.113:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 20: Voice VLAN 40 isolation test: Management VLAN 10 blocked

The Guest-VLAN isolation tests (CR14) are summarised in Figure 21, which shows Guest access to the Administration, Packhouse, Voice and Management networks blocked as intended. Individual per-destination captures are listed in Appendix A.

```
C:\>ping 192.168.26.65

Pinging 192.168.26.65 with 32 bytes of data:

Reply from 192.168.26.97: Destination host unreachable.
Reply from 192.168.26.97: Destination host unreachable.
Reply from 192.168.26.97: Destination host unreachable.
Reply from 192.168.26.97: Destination host unreachable.

Ping statistics for 192.168.26.65:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 21: Guest VLAN 50 internal VLAN isolation test: access blocked as intended

6.4 Troubleshooting performed during implementation

Table 8: Troubleshooting log

Issue	Cause	Resolution
Trunk mode rejected on CoreSwitch (“trunk encapsulation is ‘Auto’ can not be configured to ‘trunk’ mode”)	The Cisco 3560 requires the trunk encapsulation to be set before trunk mode is accepted	switchport trunk encapsulation dot1q entered first, followed by switchport mode trunk; trunks verified with show interfaces trunk

Issue	Cause	Resolution
Server-based DHCP leases lacked default gateway and DNS (0.0.0.0)	Behaviour of the server-based DHCP service in Packet Tracer; disabling and re-enabling the service and renewing the client did not change the result	DHCP service on Server0 disabled; four IOS DHCP pools created on CoreSwitch; complete leases then obtained
Client received APIPA address after ACL activation	DHCP DISCOVER (source 0.0.0.0) did not match any subnet-based permit and was discarded by the implicit deny	UDP BOOTP/DHCP permit entries added at the head of each ACL (Section 5.4)
show running-config interface vlan 40 returned an invalid-input error	Command syntax not supported by the Packet Tracer IOS image	Configuration verified with show running-config, where VLAN 40 and the inbound VOICE_ISOLATION application were visible
Guest test endpoint did not behave as a Guest client	The temporary test PC had been moved between VLANs	CoreSwitch Fa0/7 explicitly configured as an access port in VLAN 50 and verified with show interfaces status before DHCP and ACL tests were repeated
Management deny entry matched more than the Management subnet	Wildcard mask 0.0.0.15 matches 192.168.26.112 – .127, which includes the router link (192.168.26.120/30), whereas Management is a /29	Entry changed to wildcard 0.0.0.7 in both ACLs; Guest and Voice isolation tests and the match counters were repeated

6.5 Internet reachability limitation

The internal network could reach Router0 at 192.168.26.121, but a ping to 8.8.8.8 returned “Destination host unreachable” from Router0. The routing table of Router0 contained the internal static routes but no default route (gateway of last resort) and no simulated upstream network. Although the Phase 1 design depicted the Internet as an external cloud beyond the edge router, no ISP or Cloud device was implemented in the Packet Tracer topology, so no upstream path existed. For this reason, Guest-to-Internet connectivity (and the corresponding permit behaviour of GUEST_ISOLATION toward external destinations) is documented as NOT VERIFIED and is not reported as a successful test.

The recorded evidence supports part of CR14: a Guest endpoint obtained a lease, and the documented tests show blocked attempts toward protected internal VLANs. The final ACL permit allows traffic to destinations not covered by the deny entries, but external reachability has not been demonstrated. Completing that verification requires a simulated ISP/external network, an appropriate Router0 outside interface, NAT/PAT if required by the design, a default route, and successful tests to a reachable external endpoint.

7. Conclusion

The Phase 1 design was implemented in Cisco Packet Tracer with five VLANs and SVIs, two configured 802.1Q trunks, four DHCP pools, and a routed /30 connection between CoreSwitch and Router0. The report records successful DHCP assignments and selected permitted connectivity tests. Phone ports were configured with a data VLAN and Voice VLAN 40, providing VLAN-level separation of voice and data traffic. The final submission must include matching screenshots for each result claimed.

The report records GUEST_ISOLATION and VOICE_ISOLATION as blocking tested traffic toward protected VLANs while DHCP and gateway tests succeeded. Non-zero deny counters, when shown alongside the relevant test attempts, support attribution of those blocked results to the ACLs. The DHCP/ACL troubleshooting illustrates the importance of ACL ordering and the implicit deny. These conclusions depend on including the referenced, readable evidence in the submitted document.

Three limitations remain. First, Internet reachability was not verified because no upstream ISP path, NAT/PAT configuration or default route to an external network is demonstrated. Second, the ACLs are applied inbound only on the Guest and Voice SVIs; they do not enforce equivalent restrictions on traffic initiated by other VLANs. Third, the supplied screenshots include a management-endpoint configuration of 192.168.26.114/29 with gateway 192.168.26.113 and another endpoint configuration of

192.168.26.100/28 with gateway 192.168.26.97. The latter belongs to the Guest subnet, so the endpoint identity and figure caption must be checked before it is presented as Management evidence. Recommended future work is to complete external-network/NAT testing, review policy coverage on other SVIs, and validate endpoint labels against switch-port VLANs.

References

- North-West University, Department of Computer Science and Information Systems. CMPG 325 — Computer Networks: Individual Semester Project Brief, CMPG325-2026-037 (2026).
- Cisco Systems. Cisco IOS Security Configuration Guide: Access Control Lists (extended and named ACL configuration).
- Cisco Systems. Catalyst Switch Software Configuration Guide: VLANs, 802.1Q trunking, SVIs and voice VLANs.
- Cisco Systems. Cisco IOS DHCP Configuration Guide: DHCP server pools, excluded addresses and option 150.
- Lammle, T. CCNA Routing and Switching Study Guide: subnetting, VLSM and VLAN design.

Appendix A: Evidence File Index

The table below maps each saved evidence file to its figure in this report. Files marked “Supplementary” are retained in the Evidence folder as additional captures.

Table 9: Evidence file index

Figure	Evidence file	Description
Figure 4	01_Server0_Static_IP.png	Server0 static IP configuration
Figure 5	VLAN SVI Status Verification All VLAN Interfaces UpUp.png	SVI status, all VLANs up/up
Figure 6	Packhouse IP Phone Port Configuration – Data VLAN 30 and Voice VLAN 40.png	Packhouse phone port configuration
Figure 7	CoreSwitch Trunk VLAN Verification.png	Trunk verification
Figure 8	DHCP_PC0_VLAN20_SUCCESS.png	Administration DHCP lease
Figure 9	Packhouse VLAN 30 – Successful DHCP Address.png	Packhouse DHCP lease
Figure 10	DHCP Verification for Voice VLAN 40 – IP Phones Successfully Assigned.png	Voice DHCP lease (IP phones)
Figure 11	Voice VLAN DHCP Verification with VOICE_ISOLATION ACL Active.png	Voice DHCP with ACL active
Figure 12	Guest VLAN 50 – Successful DHCP Address Assignment.png	Guest DHCP lease
Figure 13	Admin VLAN 20 – Router0 Connectivity PASS.png	Admin to Router0 ping
Figure 14	ACL Match Counter Verification – Guest and Voice Isolation.png	show access-lists match counters
Figure 15	Test1_Admin_Gateway_PASS.png	Admin to gateway ping
Figure 16	Test2_Admin_to_Packhouse_PASS.png	Admin to Packhouse ping
Figure 17	Voice VLAN 40 Gateway Connectivity Test – Successful.png	Voice to voice gateway
Figure 18	Voice VLAN 40 ACL Isolation Verification – Admin and Packhouse Networks Blocked.png	Voice to Admin/Packhouse blocked
Figure 19	Voice VLAN 40 Isolation Test – Guest VLAN 50 Blocked.png	Voice to Guest blocked
Figure 20	Voice VLAN 40 Isolation Test – Management VLAN 10 Blocked.png	Voice to Management blocked
Figure 21	Guest VLAN 50 Isolation – Internal VLAN Access Blocked.png	Guest isolation (combined)
Supplementary	Guest VLAN 50 → Admin VLAN – Access Denied.png	Guest to Admin (single)
Supplementary	Guest VLAN 50 → Packhouse VLAN 30 – Access Blocked.png	Guest to Packhouse (single)
Supplementary	Guest VLAN 50 → Voice VLAN 40 – Access Blocked.png	Guest to Voice (single)
Supplementary	Guest VLAN 50 → Management VLAN 10 – Access Blocked.png	Guest to Management (single)
Supplementary	Router_to_PC0_VLAN20_SUCCESS.png	Router0 and PC0 connectivity (related to Figure 13)
Supplementary	Test3_Admin_to_Router_PASS.png	Admin to Router0 ping (related to Figure 13)
Supplementary	Voice VLAN 40 DHCP Address Assignment Verification.png	Voice DHCP verification (related to Figure 10)

Appendix B Packet Tracer Evidence Supplied for Revision

Figure B1. Evidence: Guest endpoint IPv4 configuration and failed ping to 192.168.26.113

Shows PC2 at 192.168.26.101/28 with gateway 192.168.26.97; ping to 192.168.26.113 returns destination host unreachable.

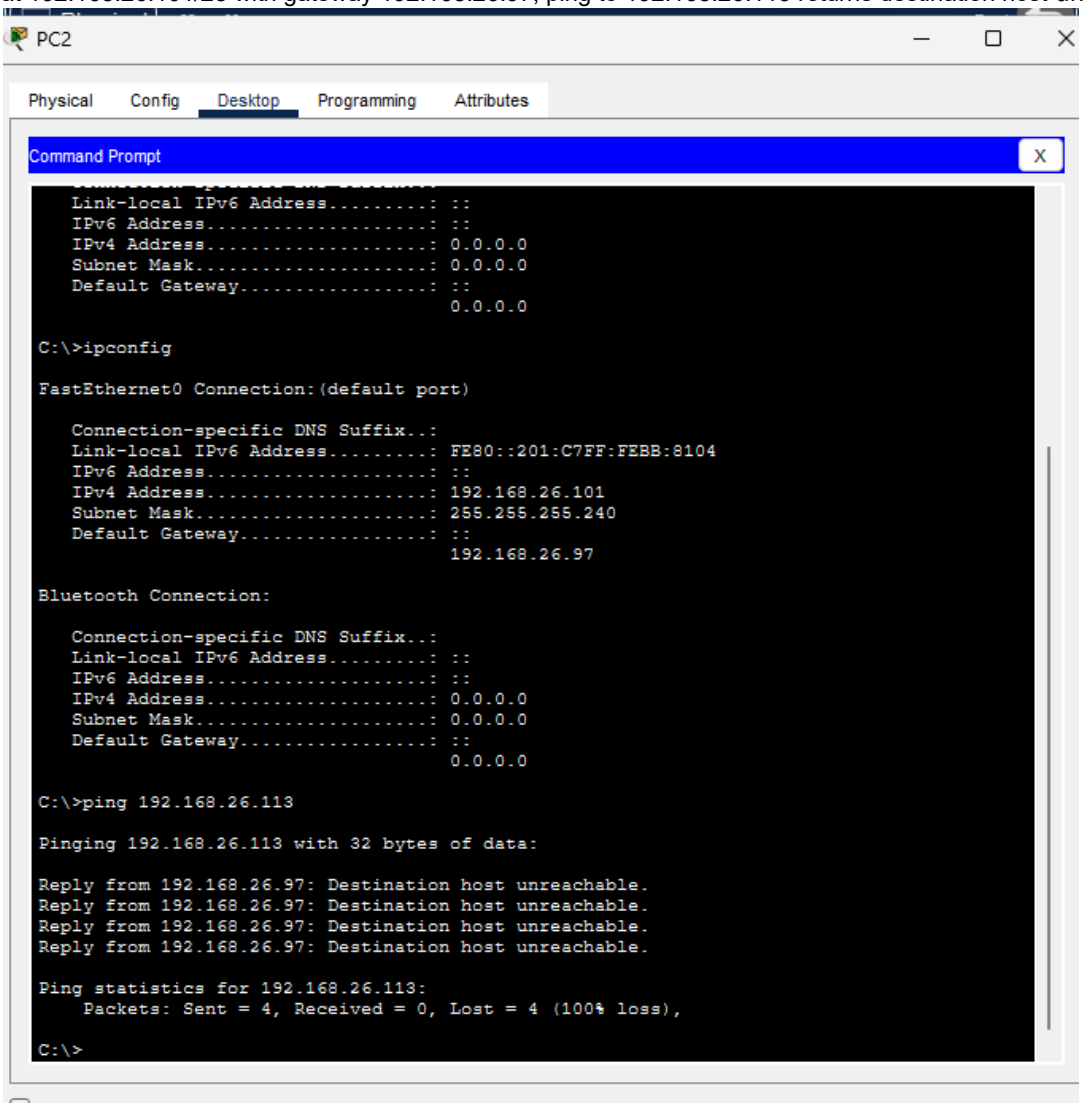


Figure B2. Evidence: Guest endpoint blocked probes to internal subnets
Shows PC2 attempting to reach 192.168.26.113 and 192.168.26.5, with destination host unreachable messages from 192.168.26.97.

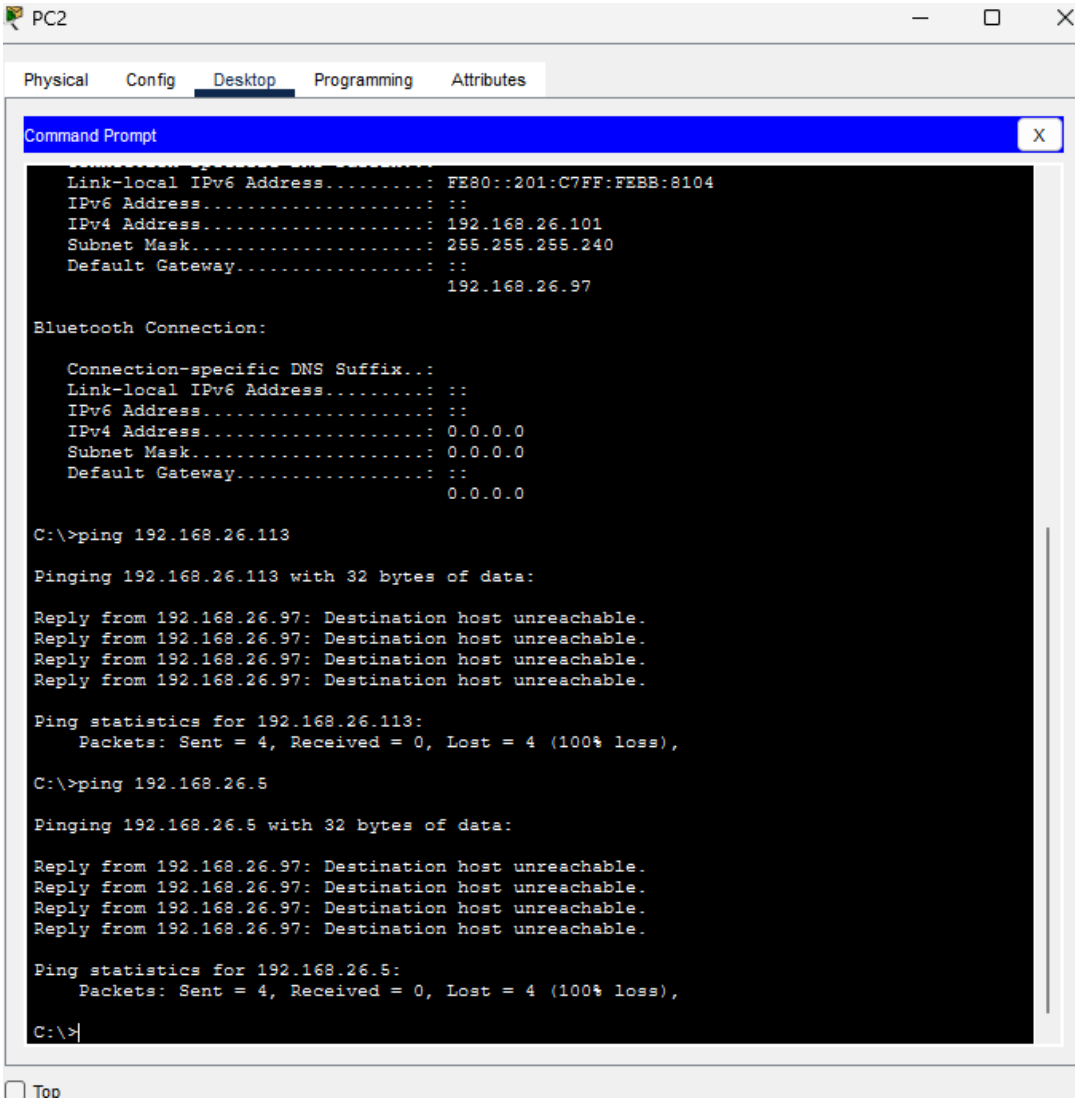


Figure B3. Evidence: Guest endpoint blocked probes to Administration and Packhouse targets

Shows failed pings to 192.168.26.5 and 192.168.26.40. This alone does not prove the ACL caused the failures; pair with ACL counters.

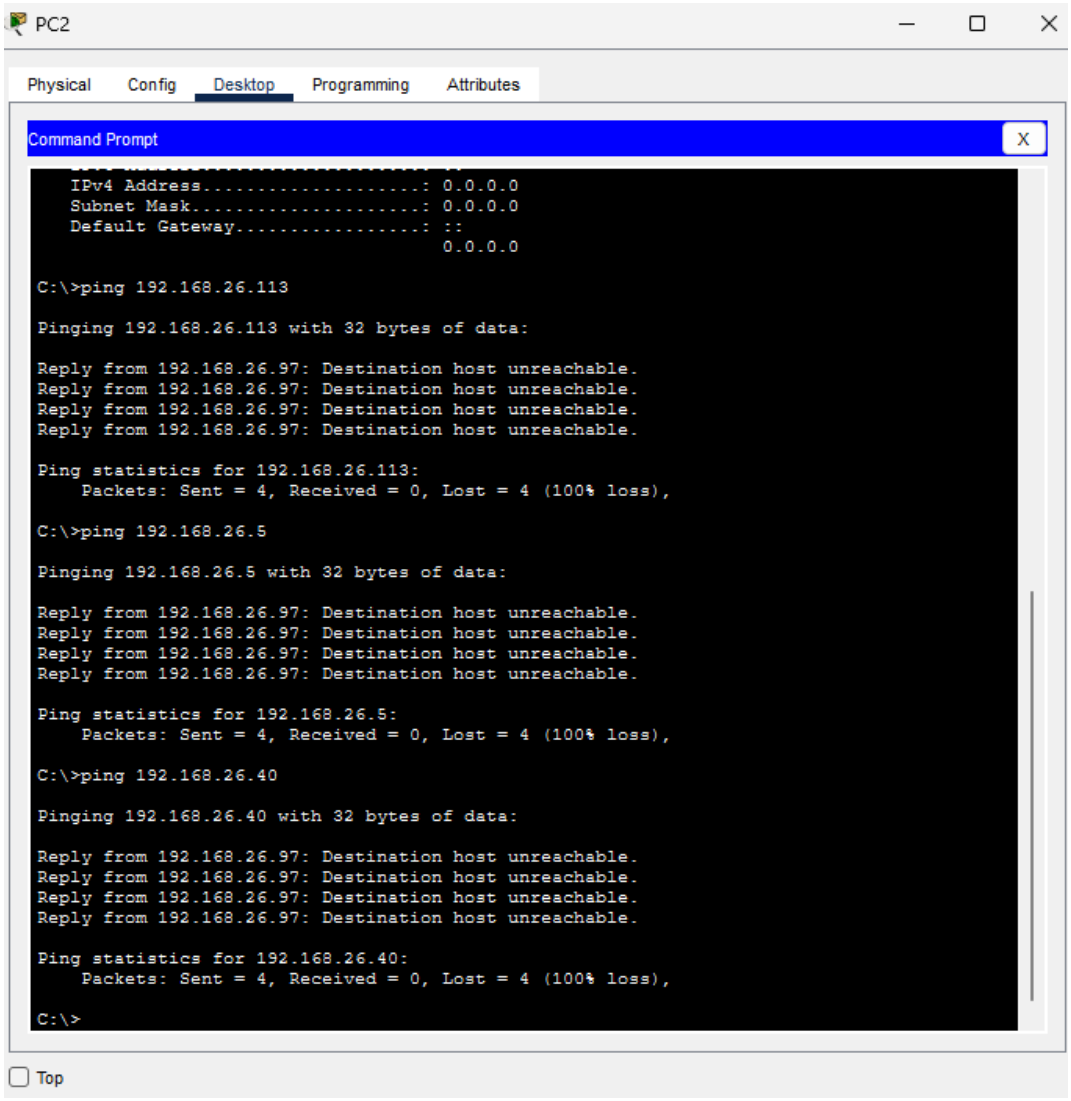


Figure B4. Evidence: Endpoint identified as Management endpoint

The displayed IPv4 configuration is 192.168.26.100/28 with gateway 192.168.26.97. This addressing belongs to the Guest subnet in the provided VLAN plan, so verify the device name/caption before treating it as Management evidence.

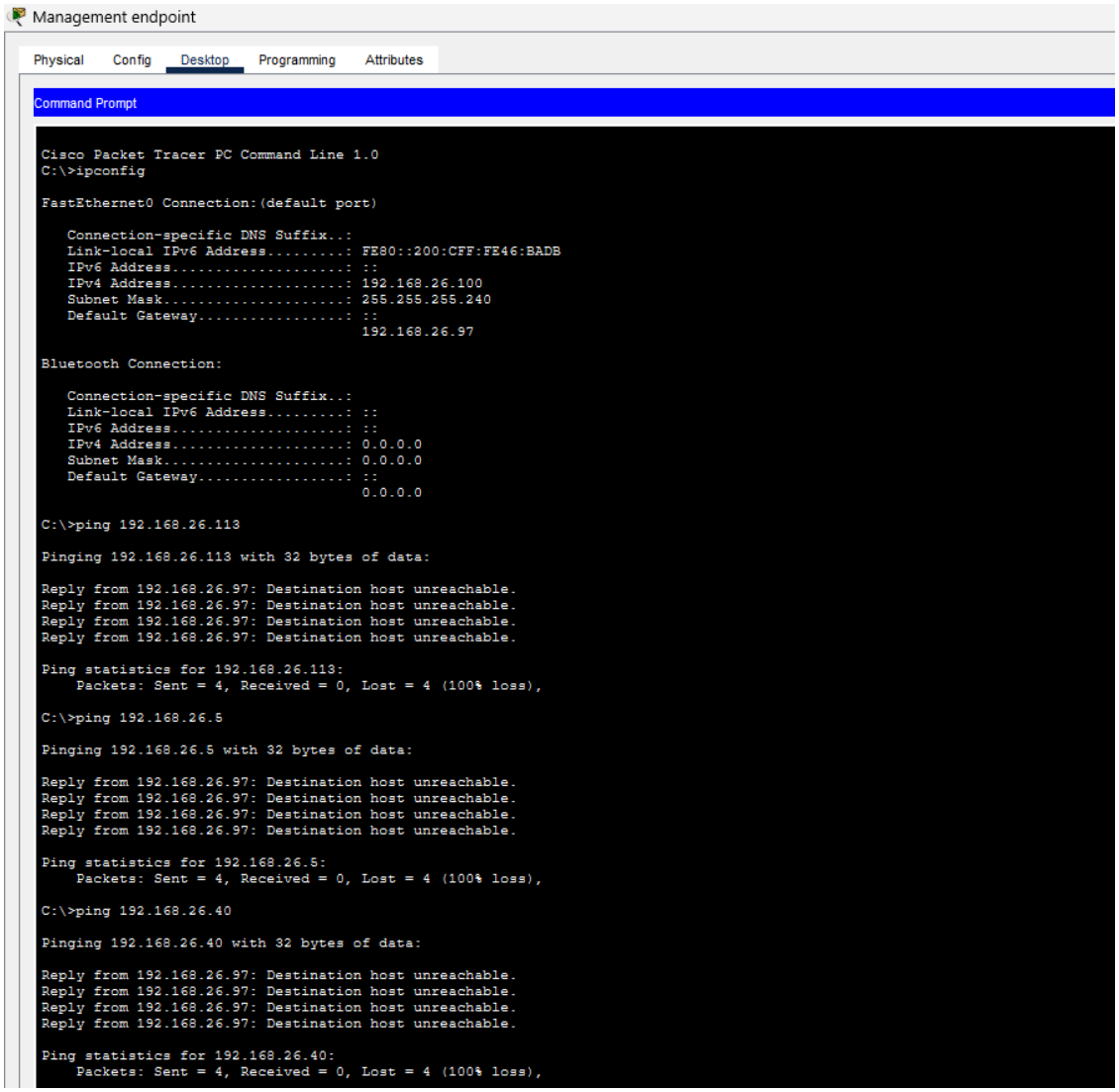


Figure B5. Evidence: Additional management-endpoint ping tests

Shows probes to 192.168.26.113, 192.168.26.5 and 192.168.26.40 returning destination host unreachable from 192.168.26.97; source is configured as 192.168.26.100/28.

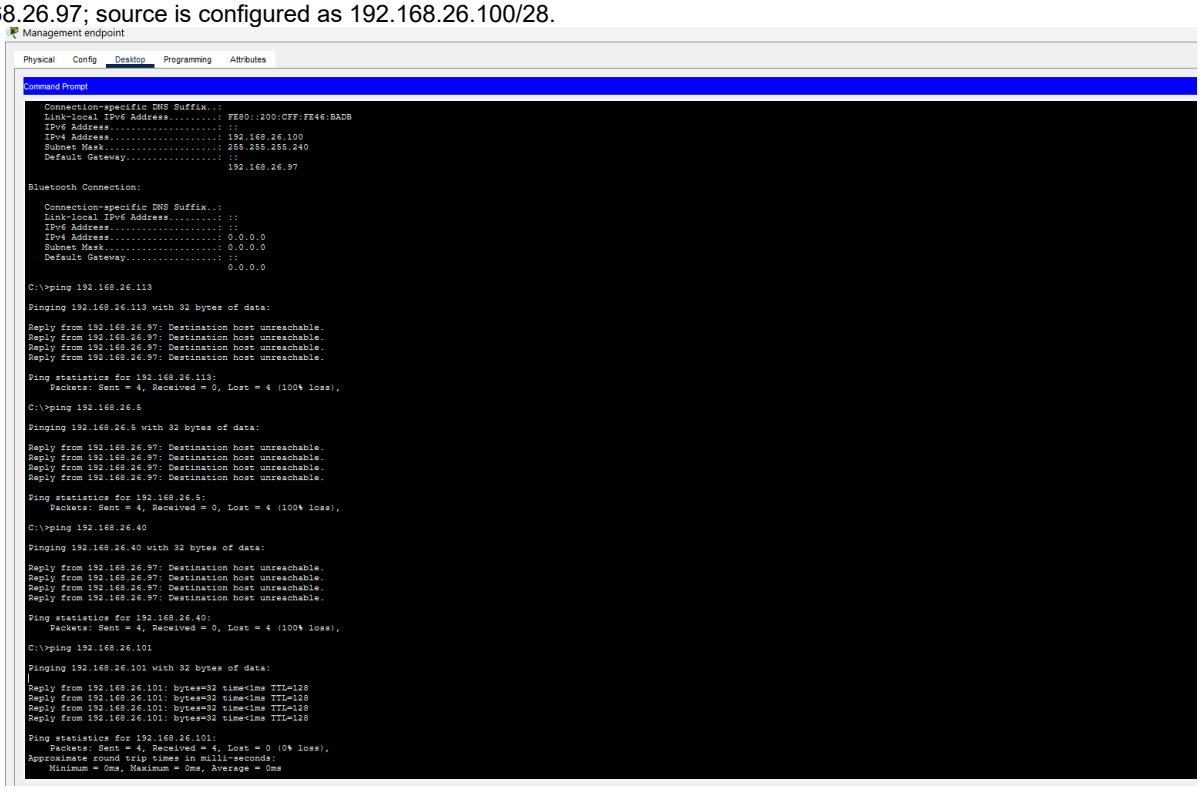


Figure B6. Evidence: PC1 in Administration VLAN reaches 192.168.26.40

PC1 is configured as 192.168.26.6/27 with gateway 192.168.26.1 and receives three replies from 192.168.26.40 after one timeout. This supports reachability, but note one packet was lost.

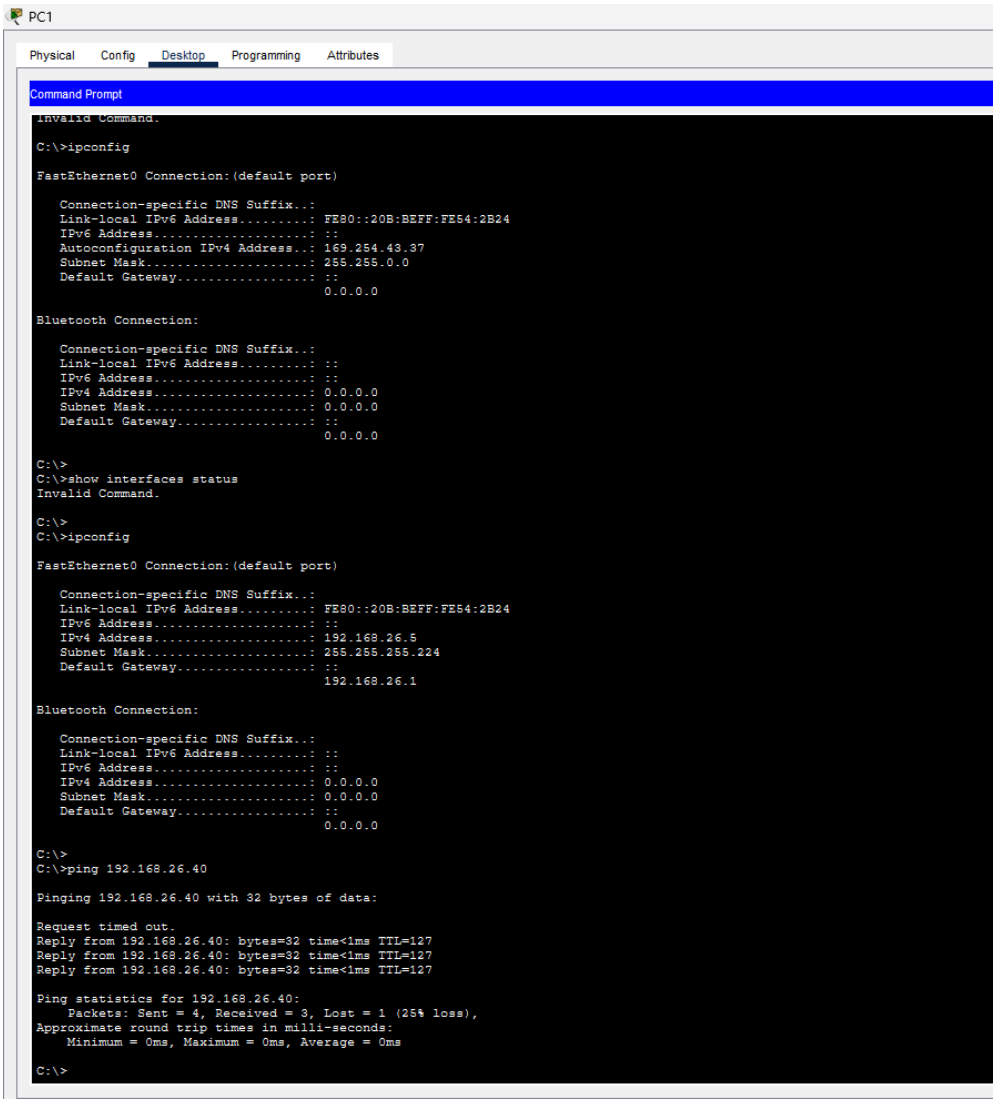


Figure B7. Evidence: Guest endpoint can reach 192.168.26.100

PC2 at 192.168.26.101/28 successfully pings 192.168.26.100 with 0% loss, confirming local subnet connectivity between these two addresses.

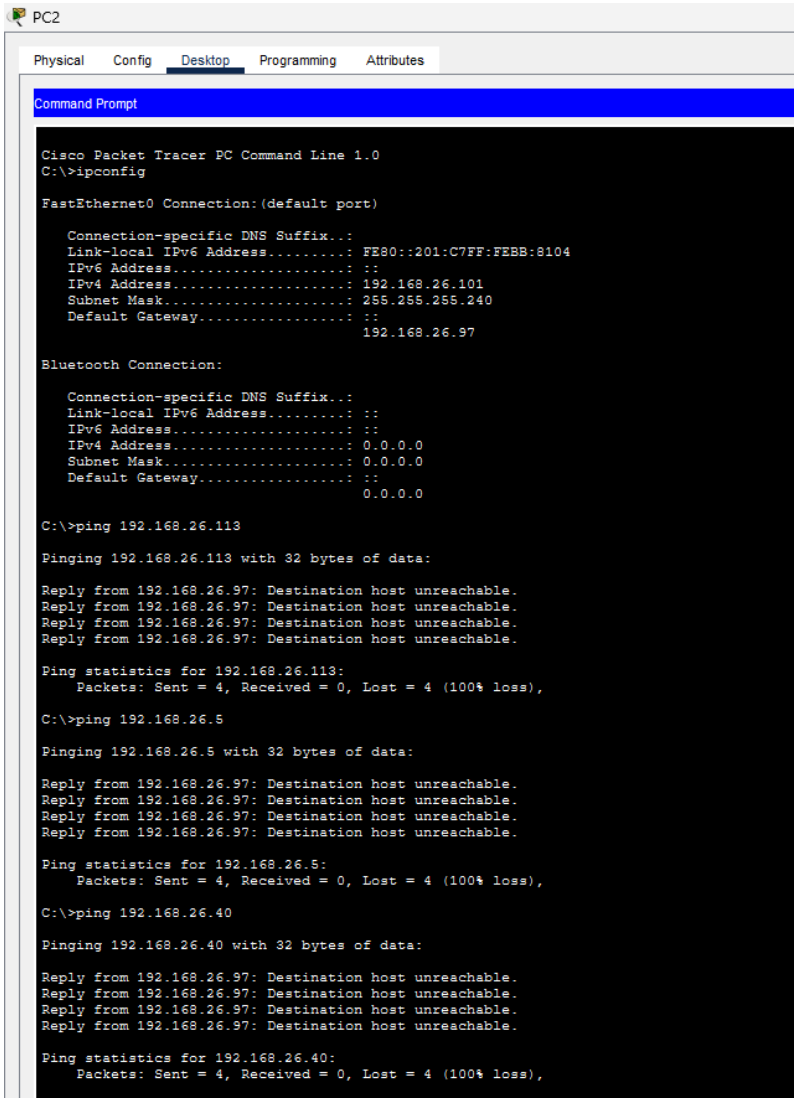
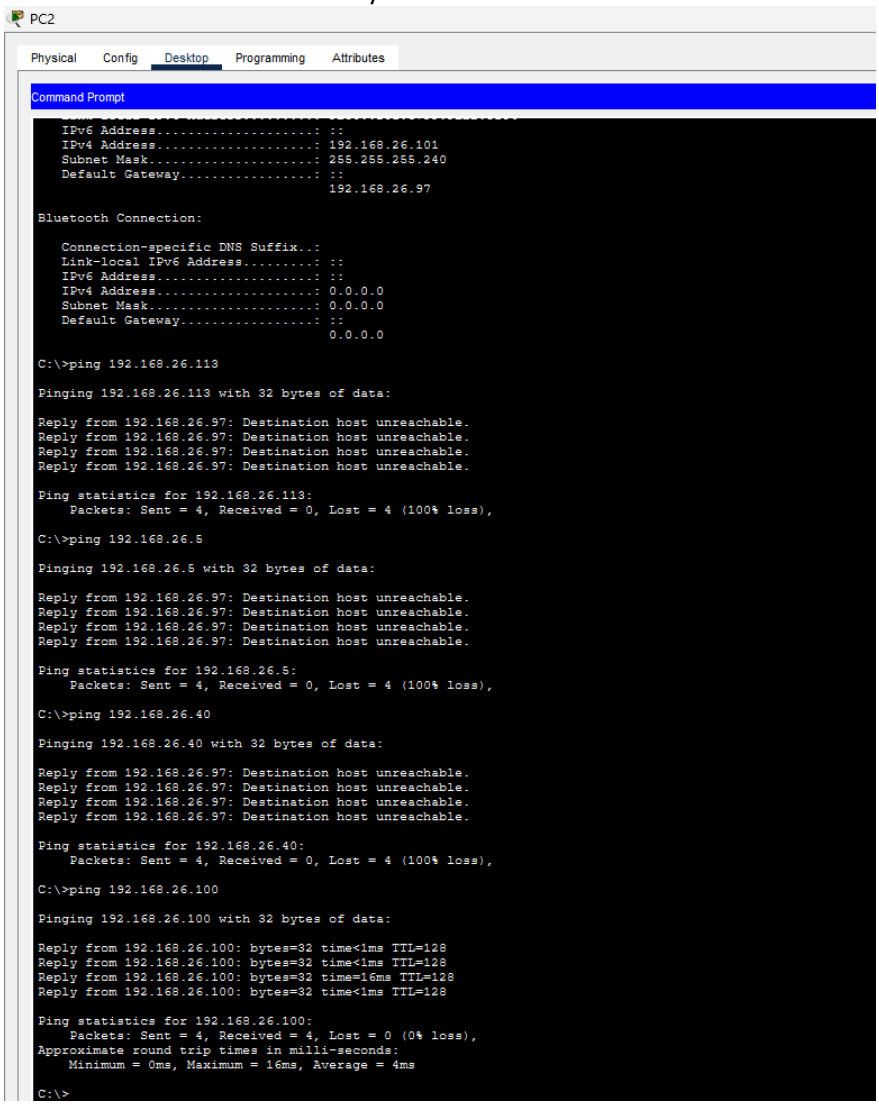


Figure B8. Evidence: Guest endpoint and gateway reachability check
Shows PC2 at 192.168.26.101/28 attempting several targets; destination host unreachable messages are returned by 192.168.26.97. Do not label as successful connectivity.



Appendix C — Additional Verification Evidence

Evidence note. These additional screenshots were selected because they show ACL configuration/counters, correct subnet addressing, and both successful and unsuccessful connectivity tests. The captions describe only visible output; unsuccessful pings are not treated as proof of ACL enforcement without matching counter evidence.

Figure C1. ACL configuration and match counters on CoreSwitch

The displayed show access-lists output includes VOICE_ISOLATION and GUEST_ISOLATION entries with match counts, and show running-config confirms both ACLs are applied inbound. This supports that the ACLs are configured and have matched traffic; correlate each counter with the corresponding endpoint test before attributing a particular failure.

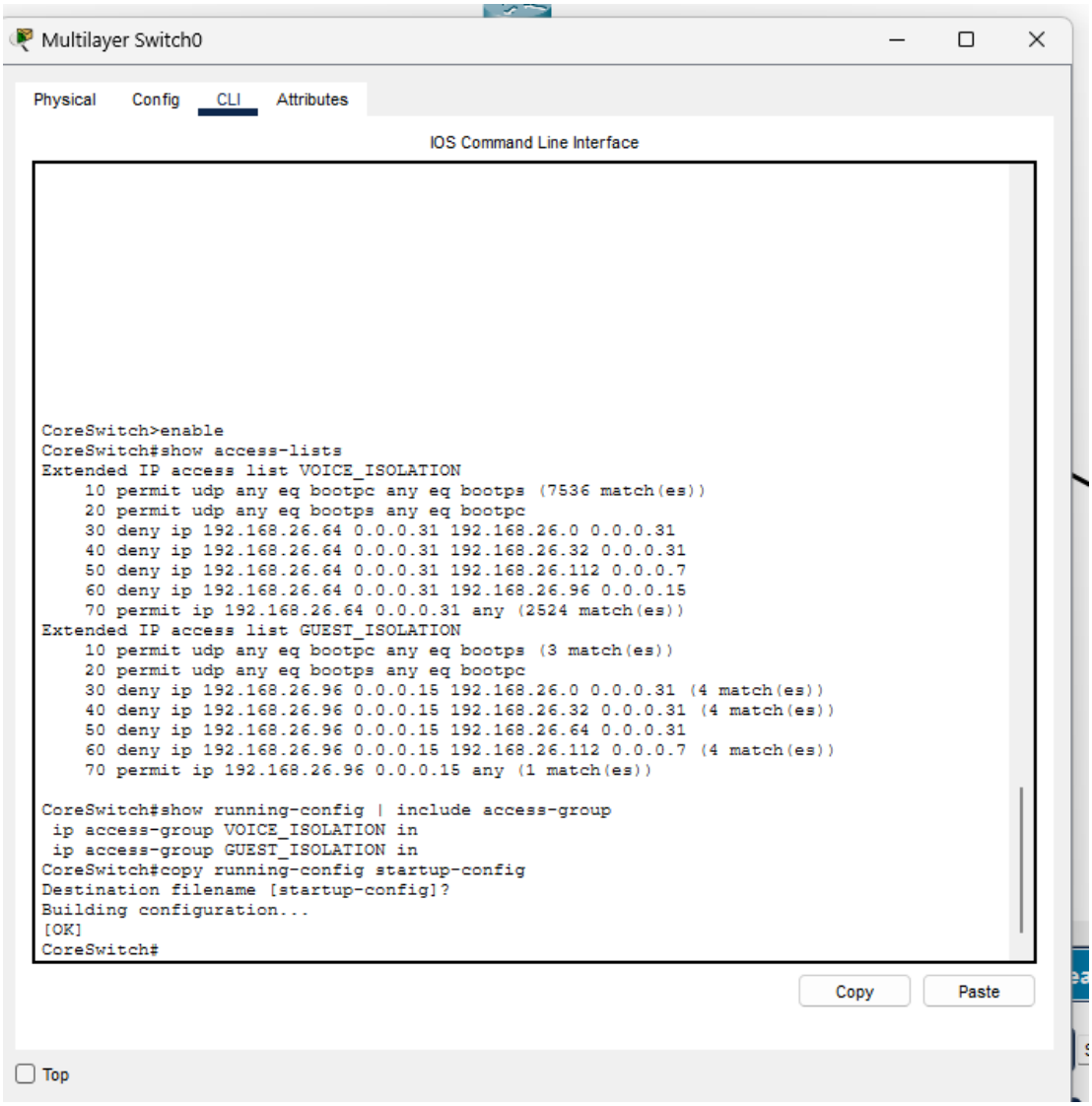


Figure C2. Management endpoint addressing and gateway test

The endpoint is configured as 192.168.26.114/29 with default gateway 192.168.26.113. The ping to 192.168.26.113 receives four replies with no packet loss. This is consistent with the Management subnet 192.168.26.112/29.

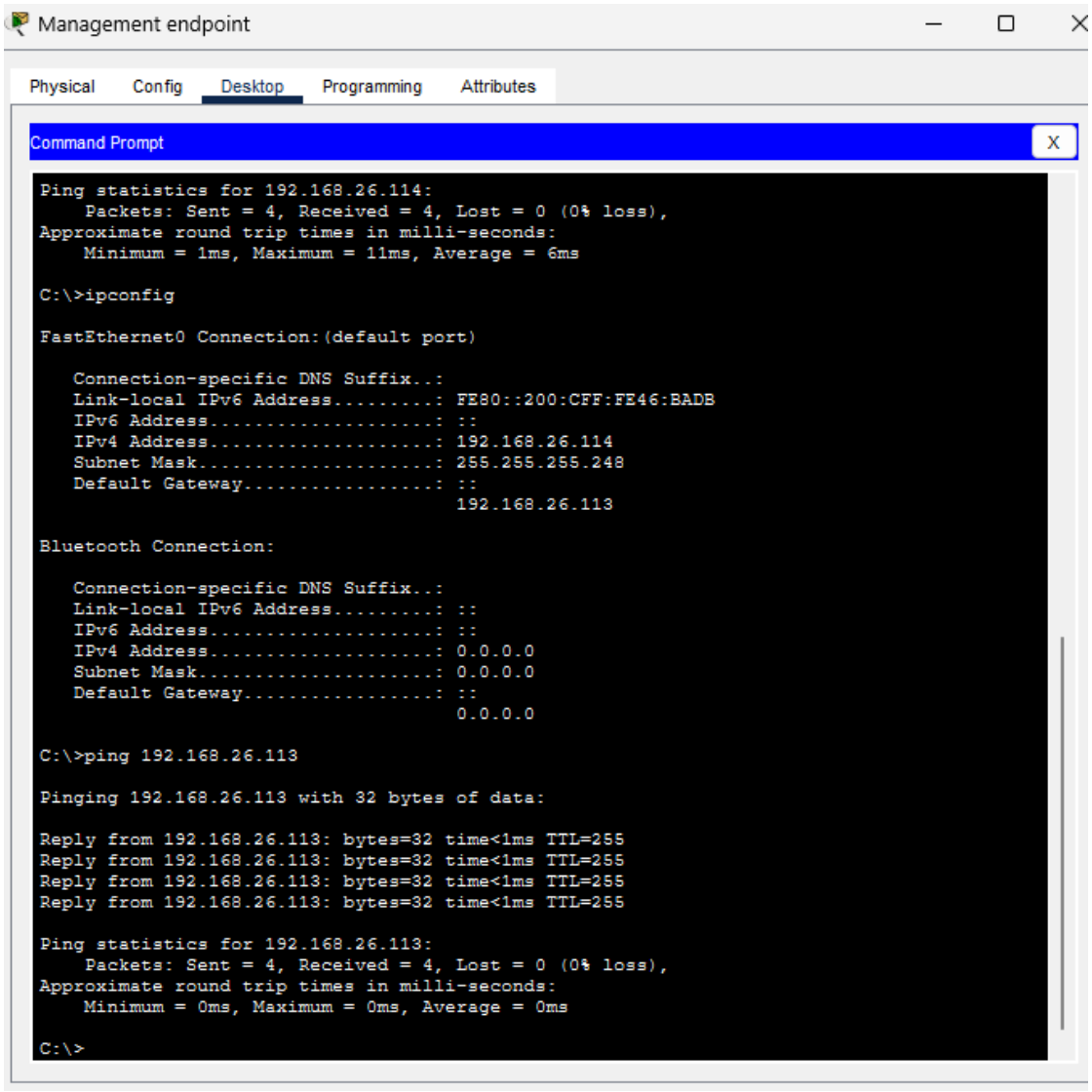

Management endpoint
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Ping statistics for 192.168.26.114:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 11ms, Average = 6ms
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::200:CFF:FE46:BADB
IPv6 Address.....: ::
IPv4 Address.....: 192.168.26.114
Subnet Mask.....: 255.255.255.248
Default Gateway.....: ::
192.168.26.113
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>ping 192.168.26.113
Pinging 192.168.26.113 with 32 bytes of data:
Reply from 192.168.26.113: bytes=32 time<1ms TTL=255
Reply from 192.168.26.113: bytes=32 time<1ms TTL=255
Reply from 192.168.26.113: bytes=32 time<1ms TTL=255
Reply from 192.168.26.113: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.26.113:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>

Figure C3. Guest endpoint: protected destinations fail while gateway responds

The PC2 screen shows attempts to reach 192.168.26.114, 192.168.26.113, 192.168.26.40 and 192.168.26.5 returning destination host unreachable from 192.168.26.97; the ping to 192.168.26.97 itself succeeds. This shows the endpoint can reach its gateway but not the tested destinations. Use the ACL counters to confirm whether ACL filtering caused the failures.

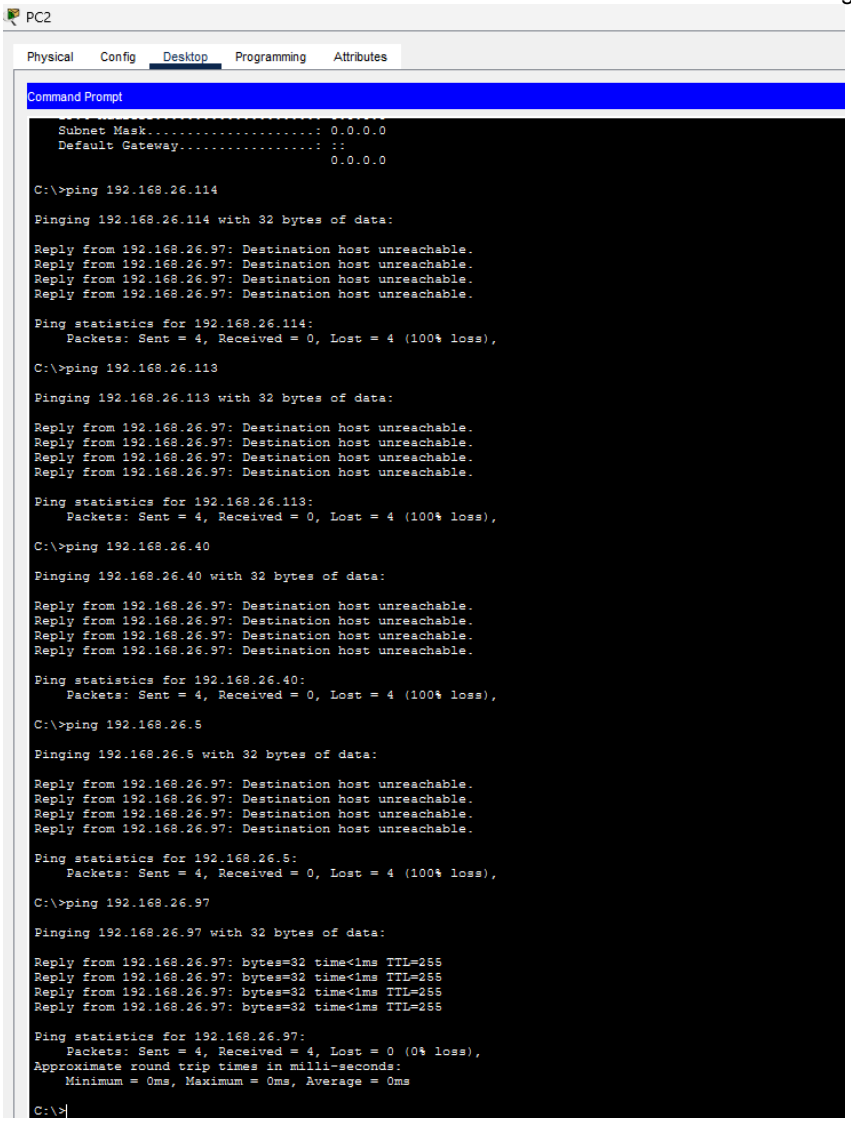


Figure C4. Administration endpoint addressing and inter-VLAN ping

PC1 is configured as 192.168.26.5/27 with gateway 192.168.26.1. The ping to 192.168.26.40 has one timeout followed by three replies (25% packet loss). Report this as partial/intermittent success, not a clean 0% loss result.

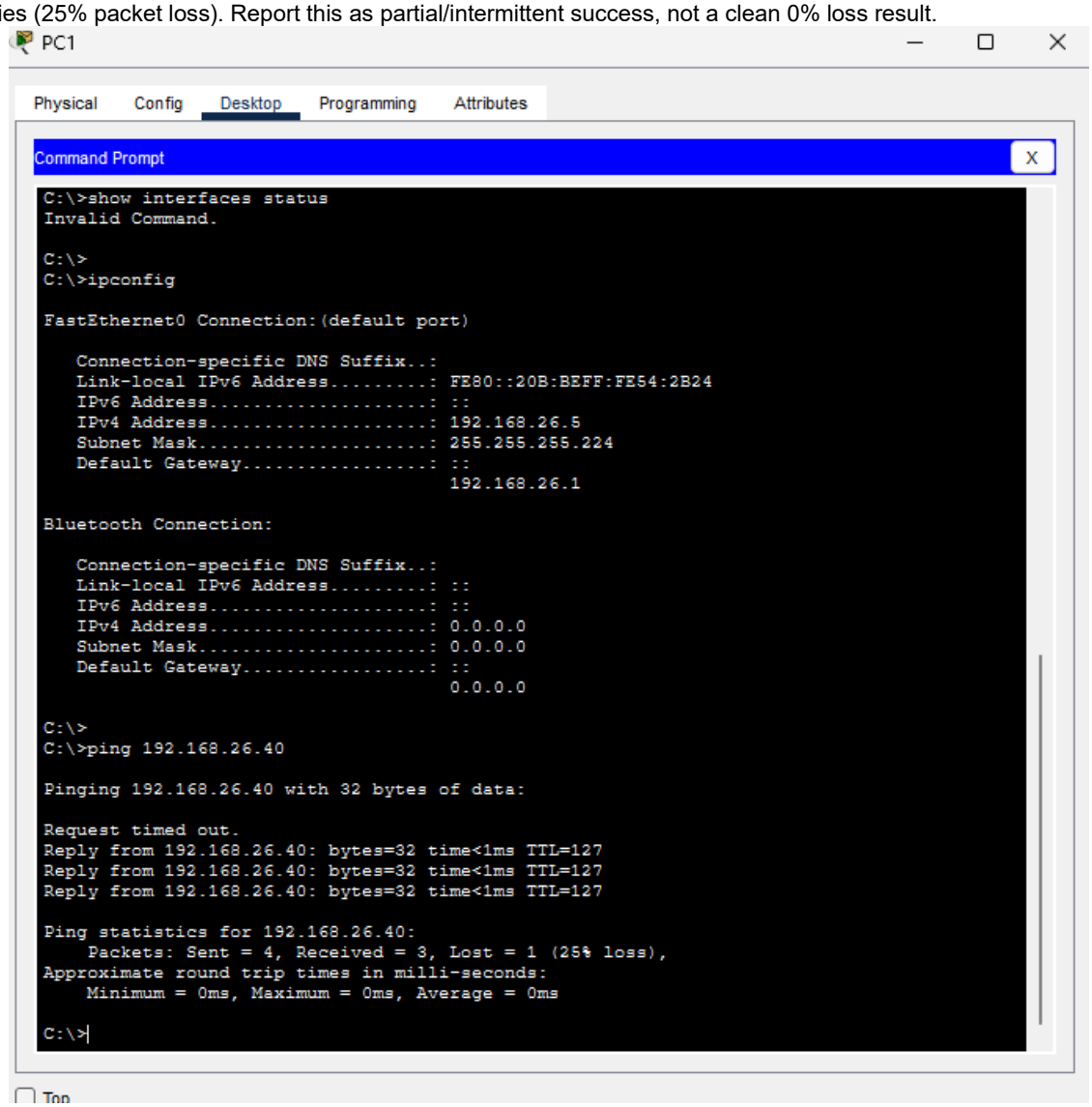


Figure C5. Guest endpoint IP configuration

The PC2 configuration shows IPv4 address 192.168.26.101, subnet mask 255.255.255.240 (/28), and gateway 192.168.26.97, consistent with the Guest subnet 192.168.26.96/28.

